

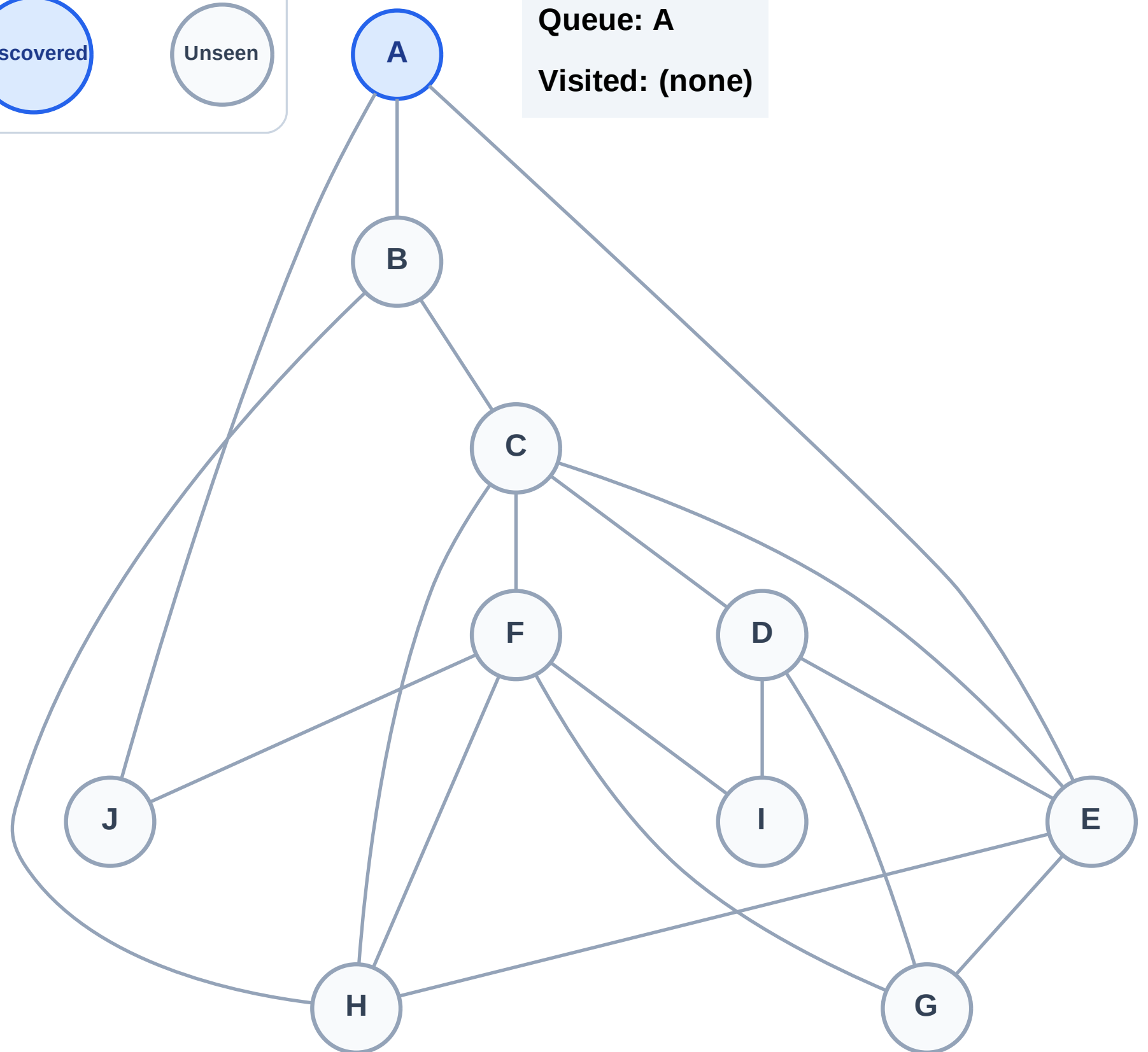
BFS Traversal

Step 1: Start at A

Legend



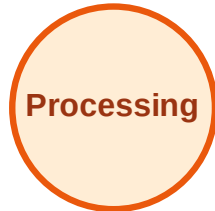
Queue: A
Visited: (none)



BFS Traversal

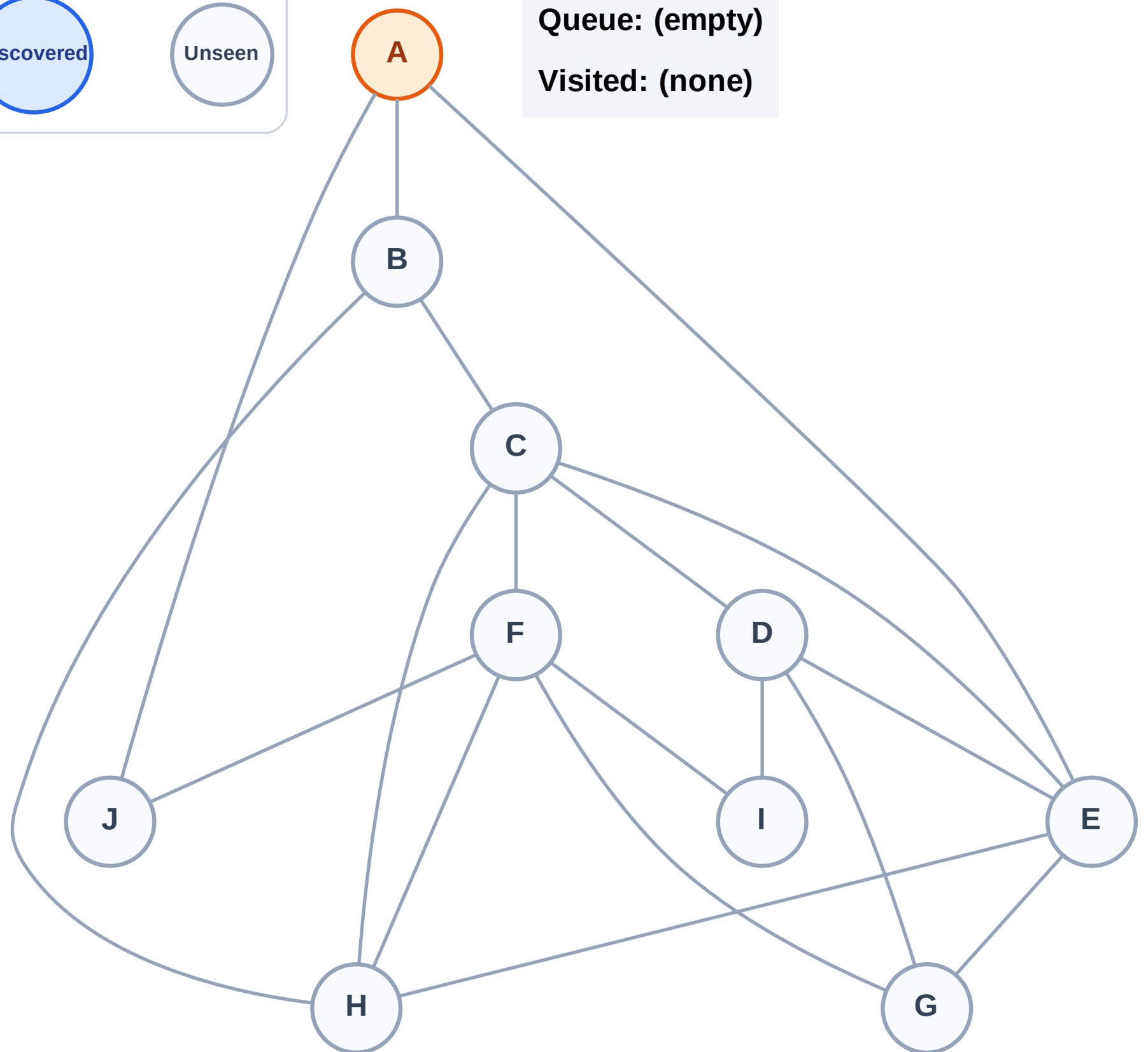
Step 2: Process node A

Legend



Queue: (empty)

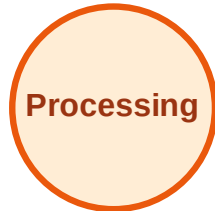
Visited: (none)



BFS Traversal

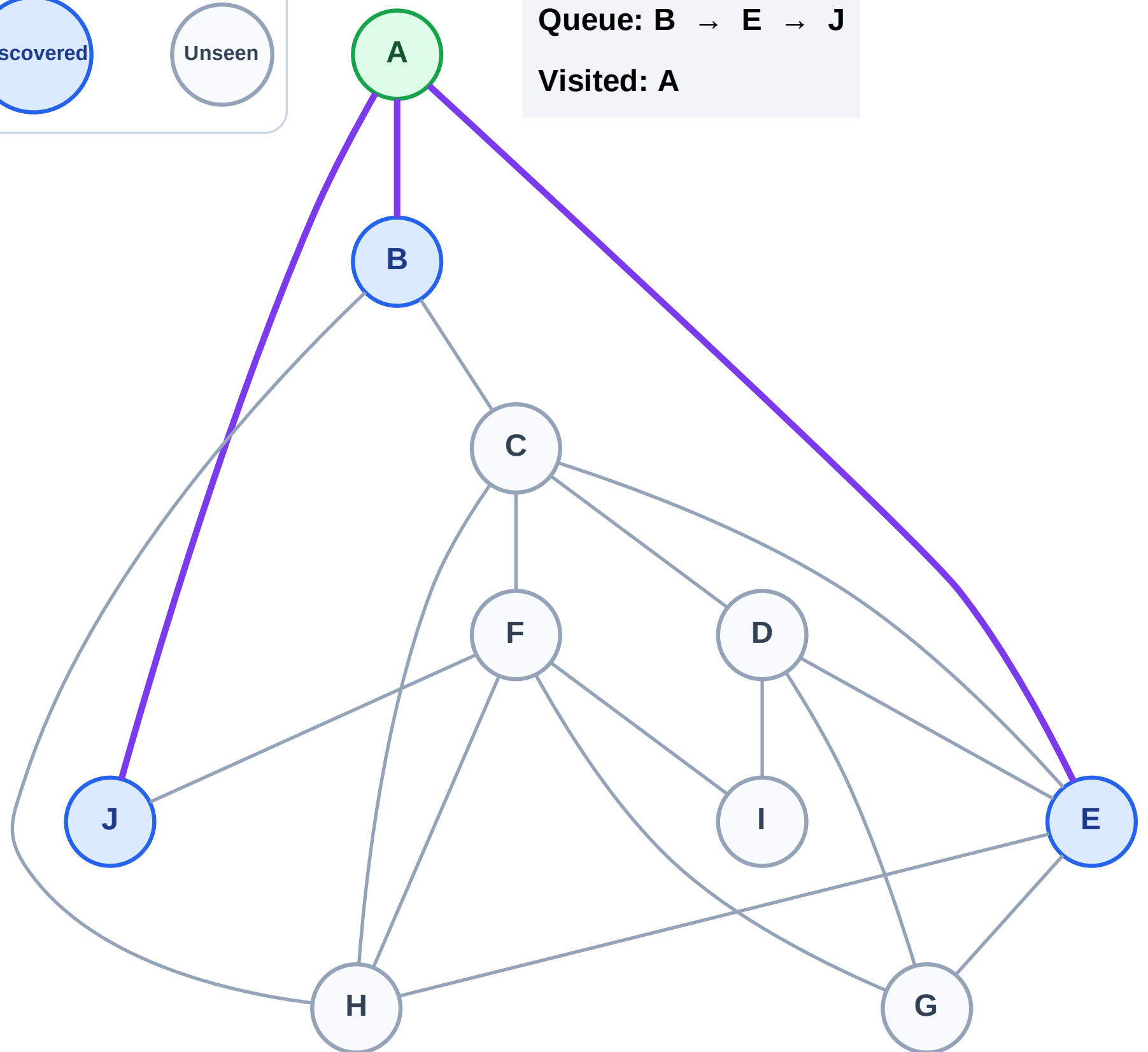
Step 3: Discover B, E, J from A

Legend



Queue: B → E → J

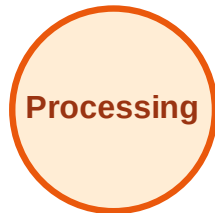
Visited: A



BFS Traversal

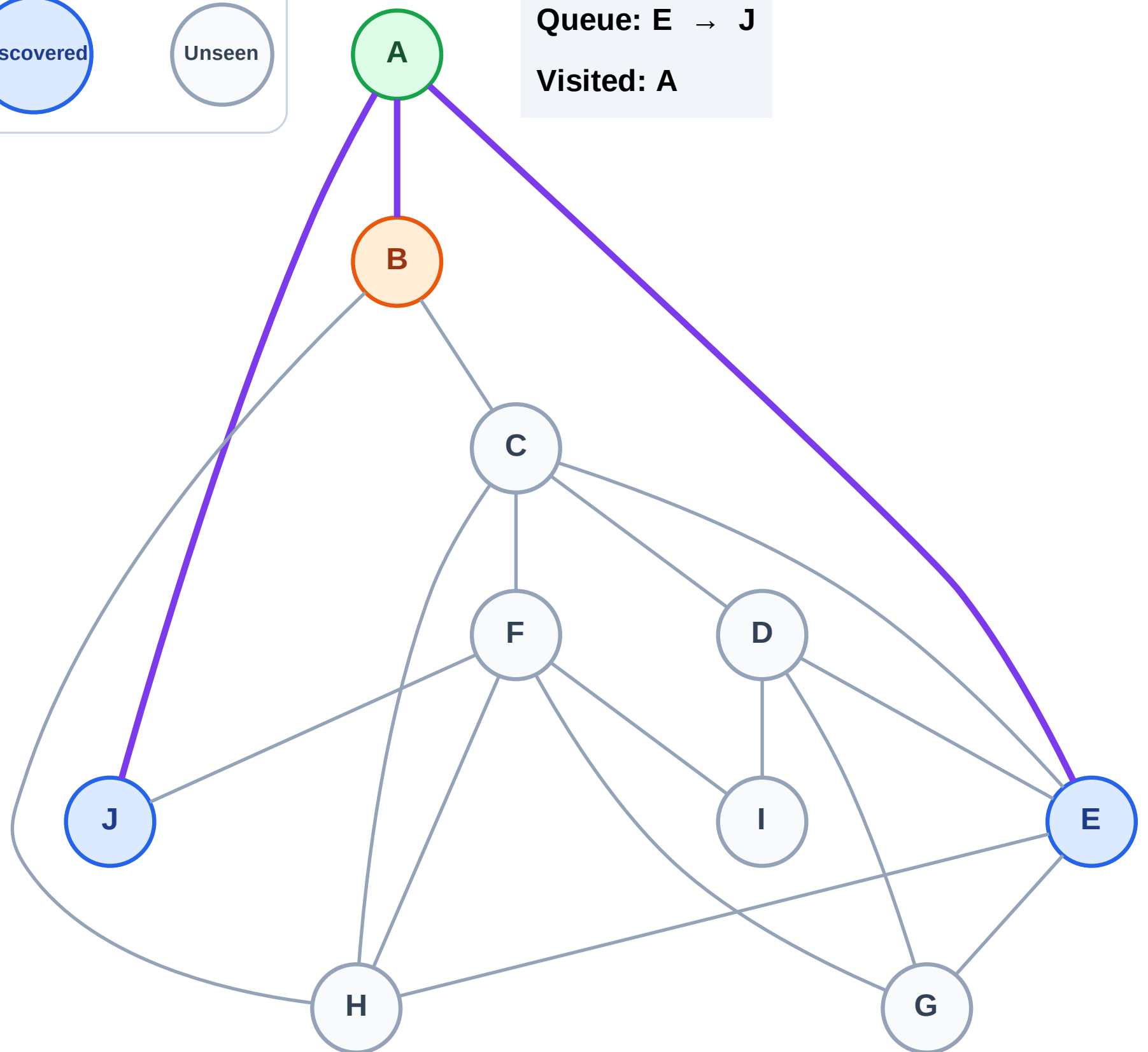
Step 4: Process node B

Legend



Queue: E → J

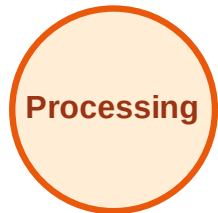
Visited: A



BFS Traversal

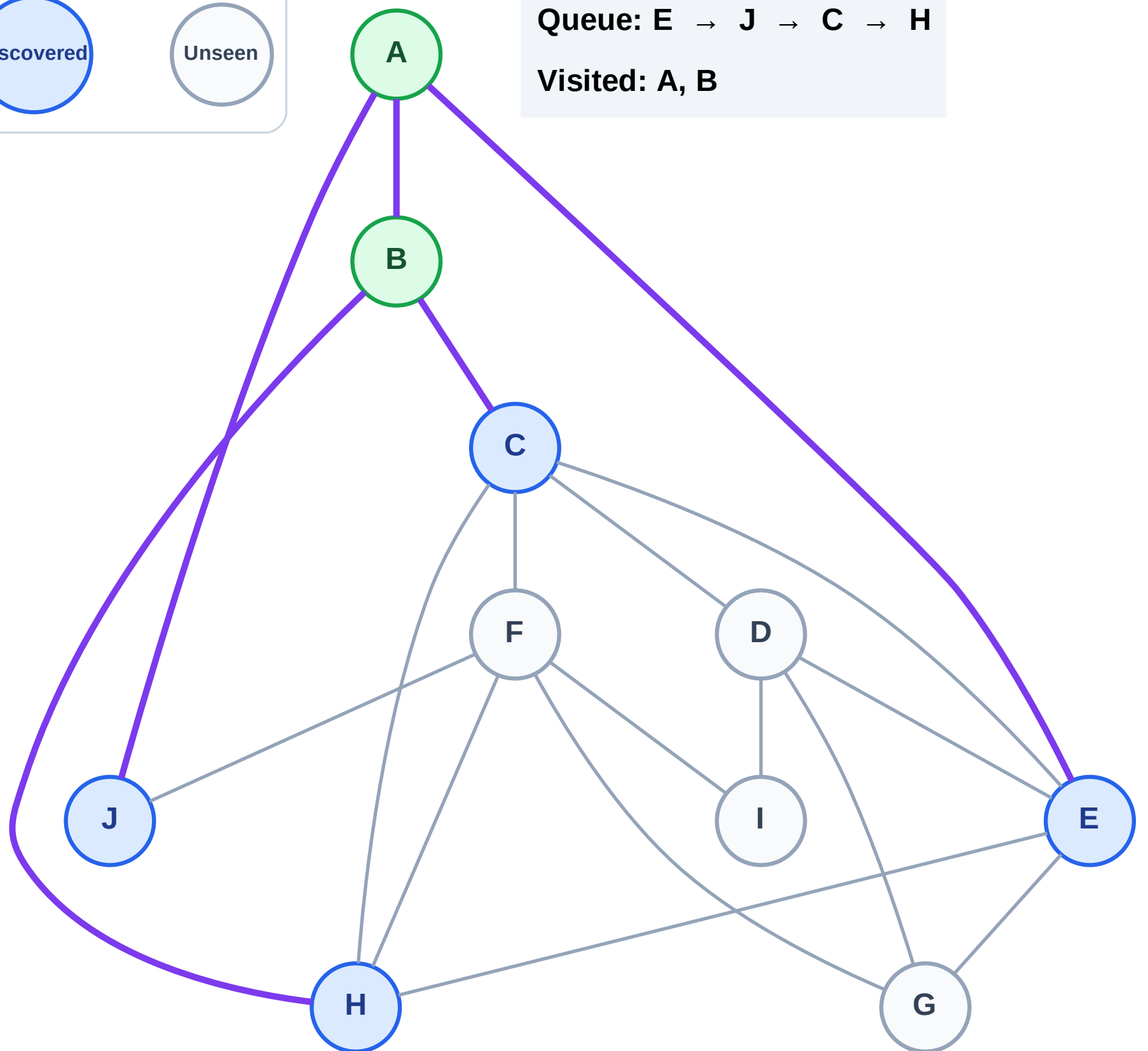
Step 5: Discover C, H from B

Legend



Queue: E → J → C → H

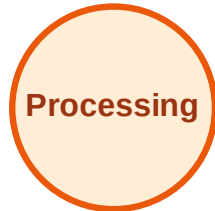
Visited: A, B



BFS Traversal

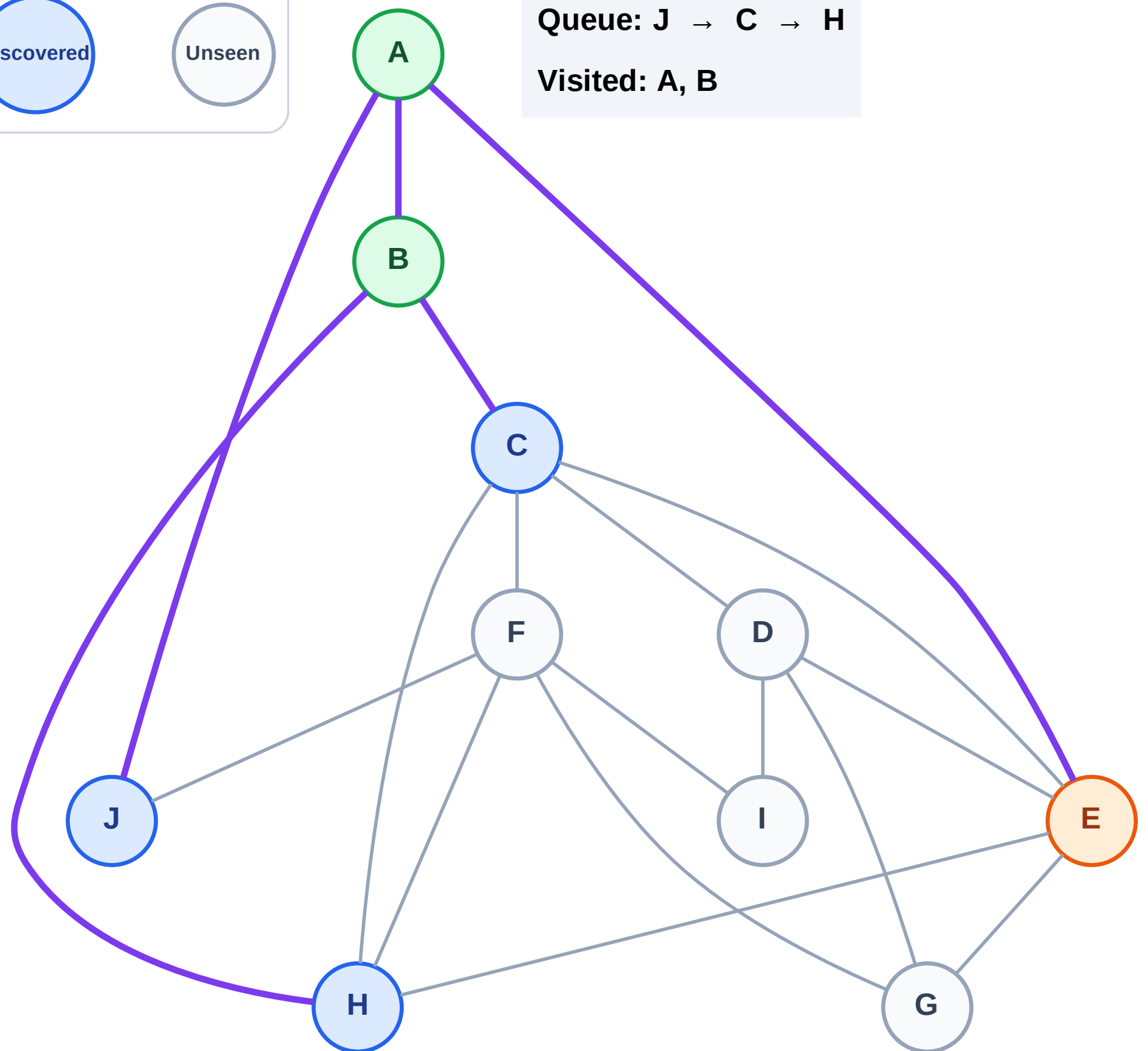
Step 6: Process node E

Legend



Queue: J → C → H

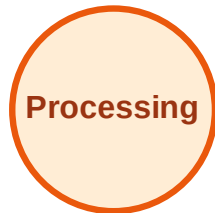
Visited: A, B



BFS Traversal

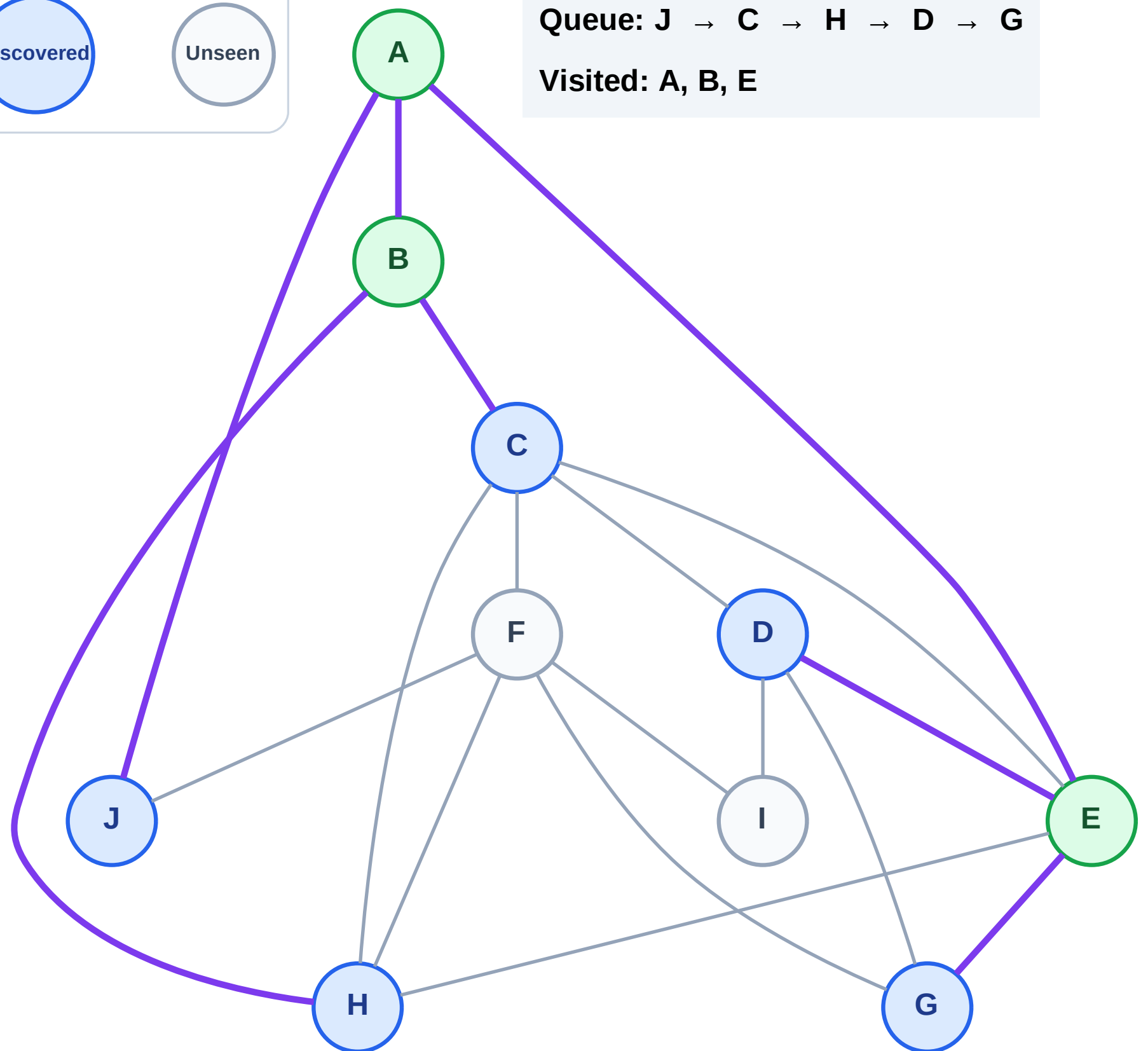
Step 7: Discover D, G from E

Legend



Queue: J → C → H → D → G

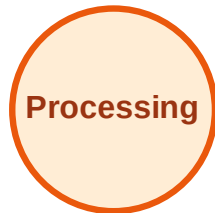
Visited: A, B, E



BFS Traversal

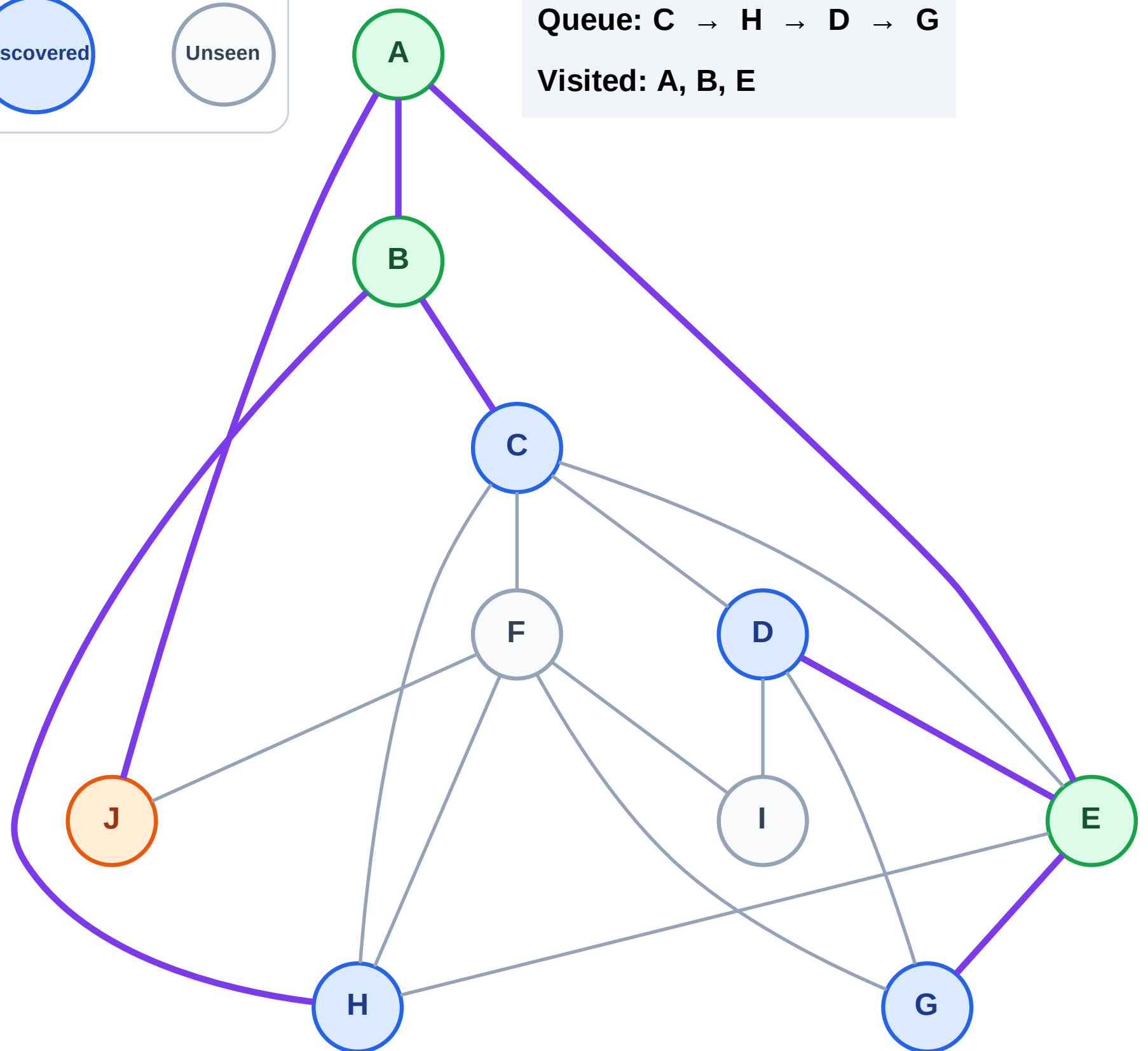
Step 8: Process node J

Legend



Queue: C → H → D → G

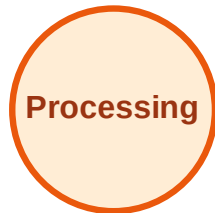
Visited: A, B, E



BFS Traversal

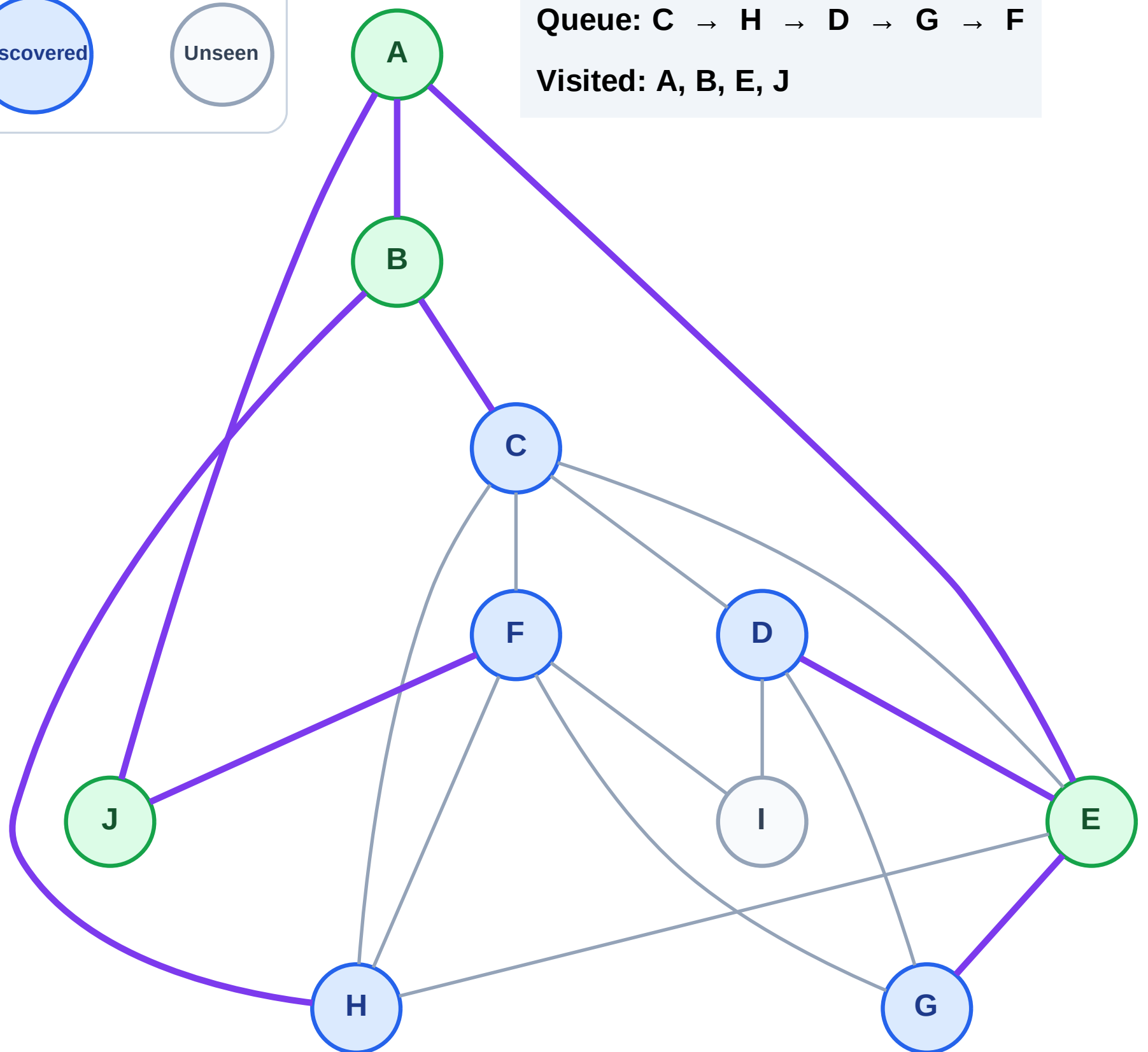
Step 9: Discover F from J

Legend



Queue: C → H → D → G → F

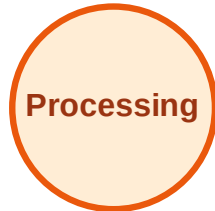
Visited: A, B, E, J



BFS Traversal

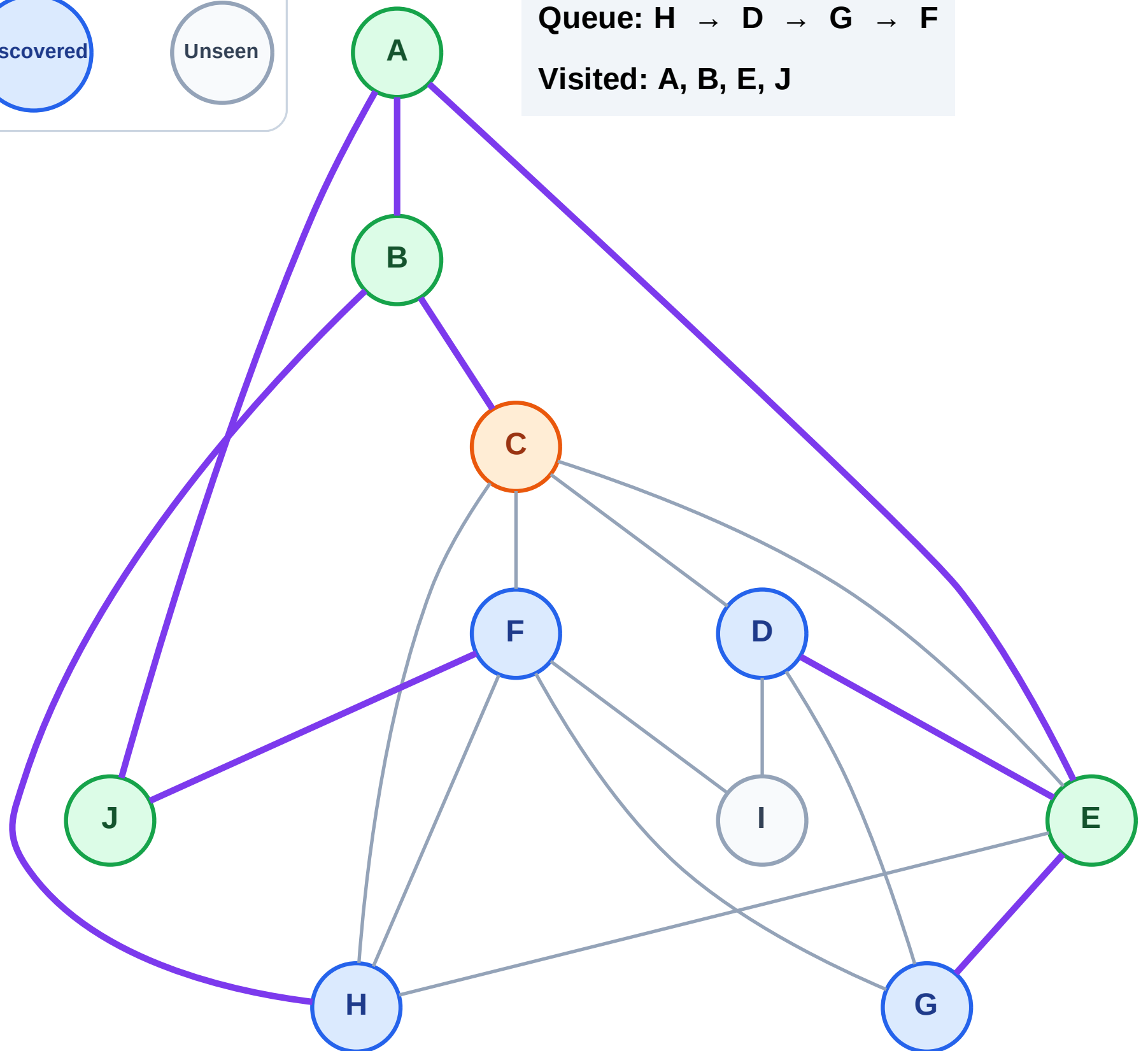
Step 10: Process node C

Legend



Queue: H → D → G → F

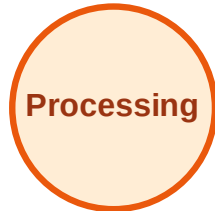
Visited: A, B, E, J



BFS Traversal

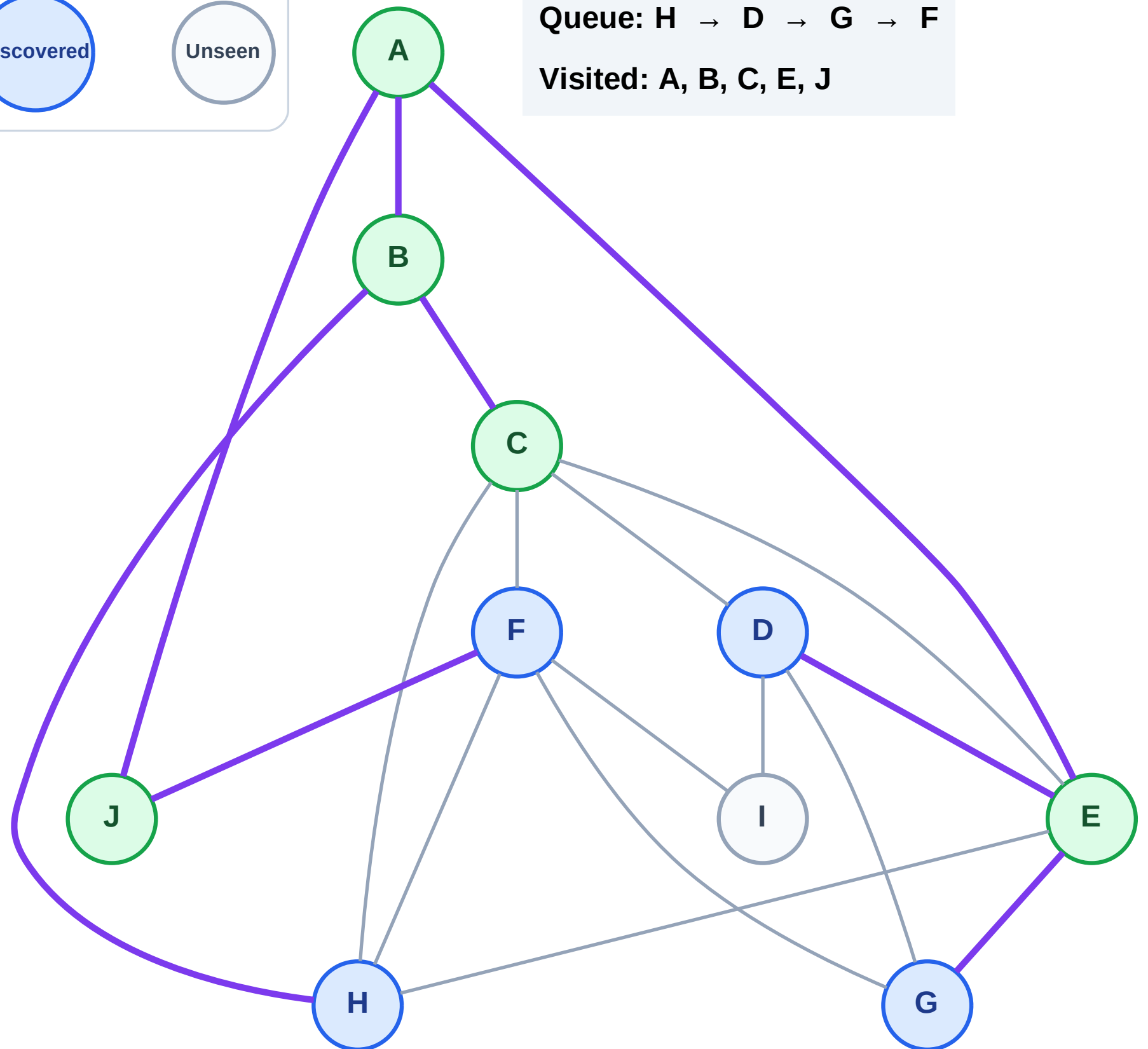
Step 11: No new neighbors from C

Legend



Queue: H → D → G → F

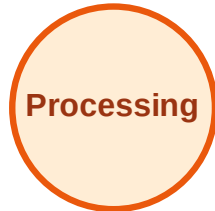
Visited: A, B, C, E, J



BFS Traversal

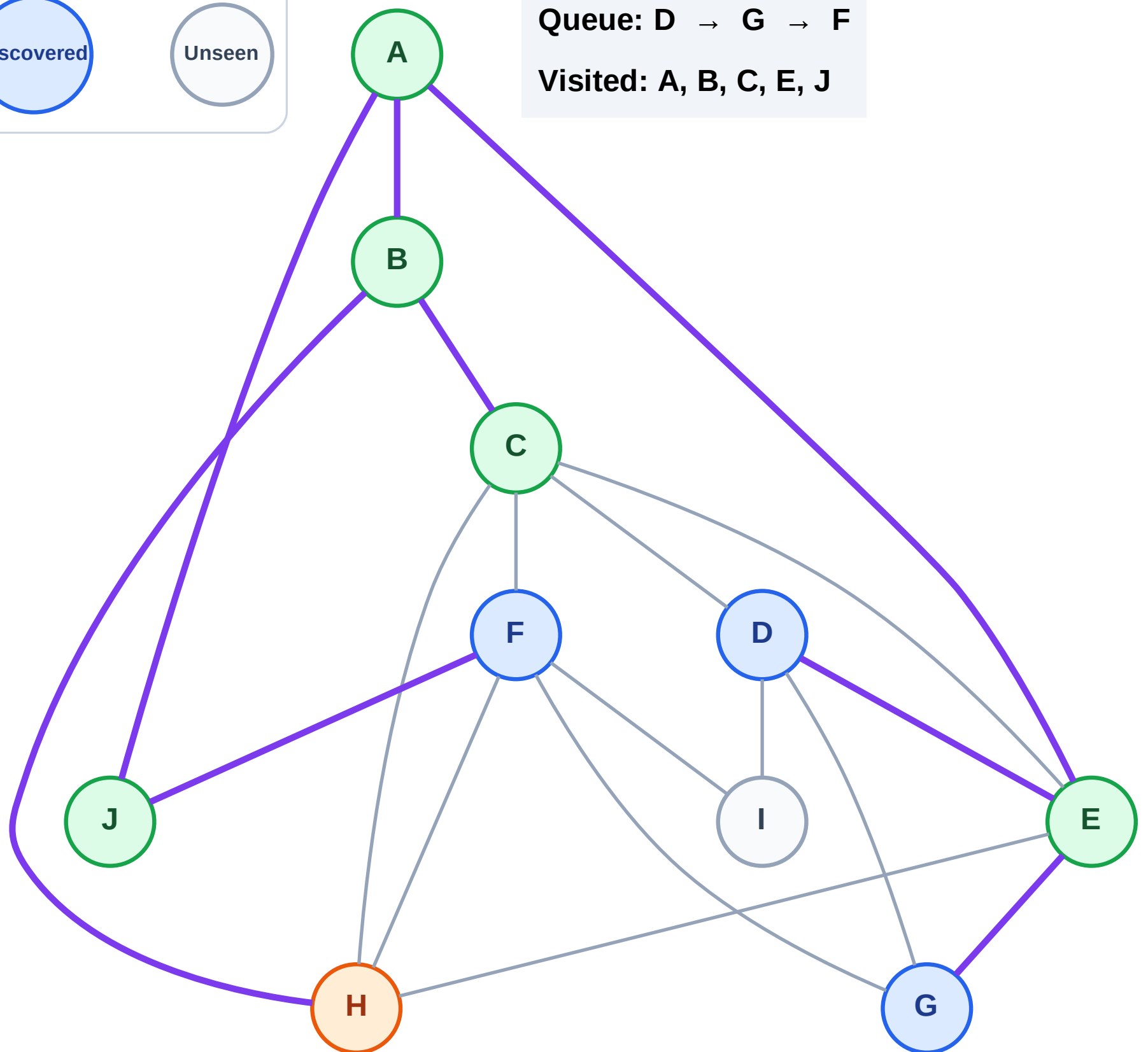
Step 12: Process node H

Legend



Queue: D → G → F

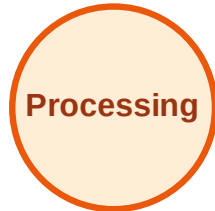
Visited: A, B, C, E, J



BFS Traversal

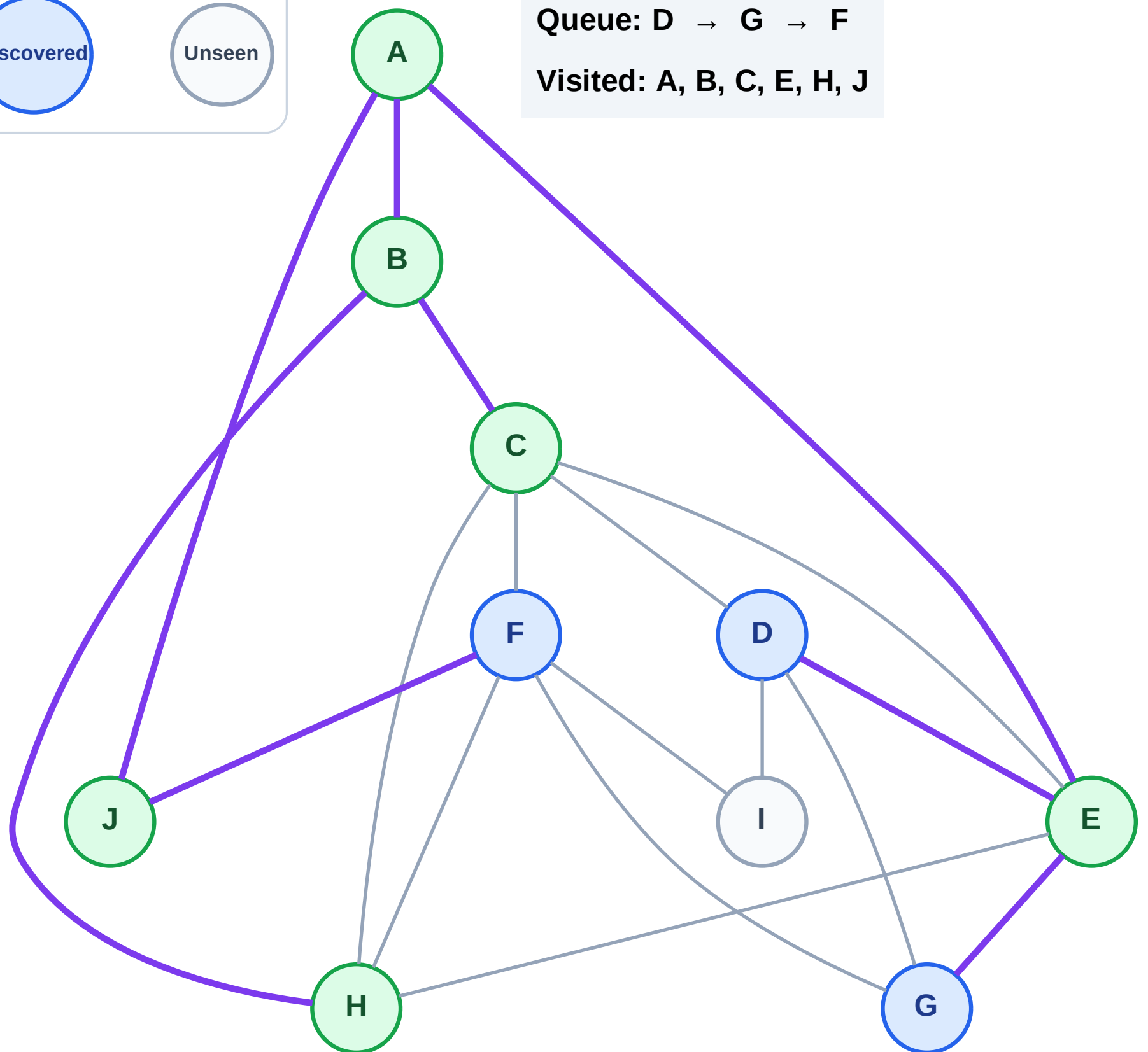
Step 13: No new neighbors from H

Legend



Queue: D → G → F

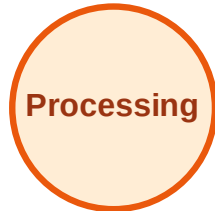
Visited: A, B, C, E, H, J



BFS Traversal

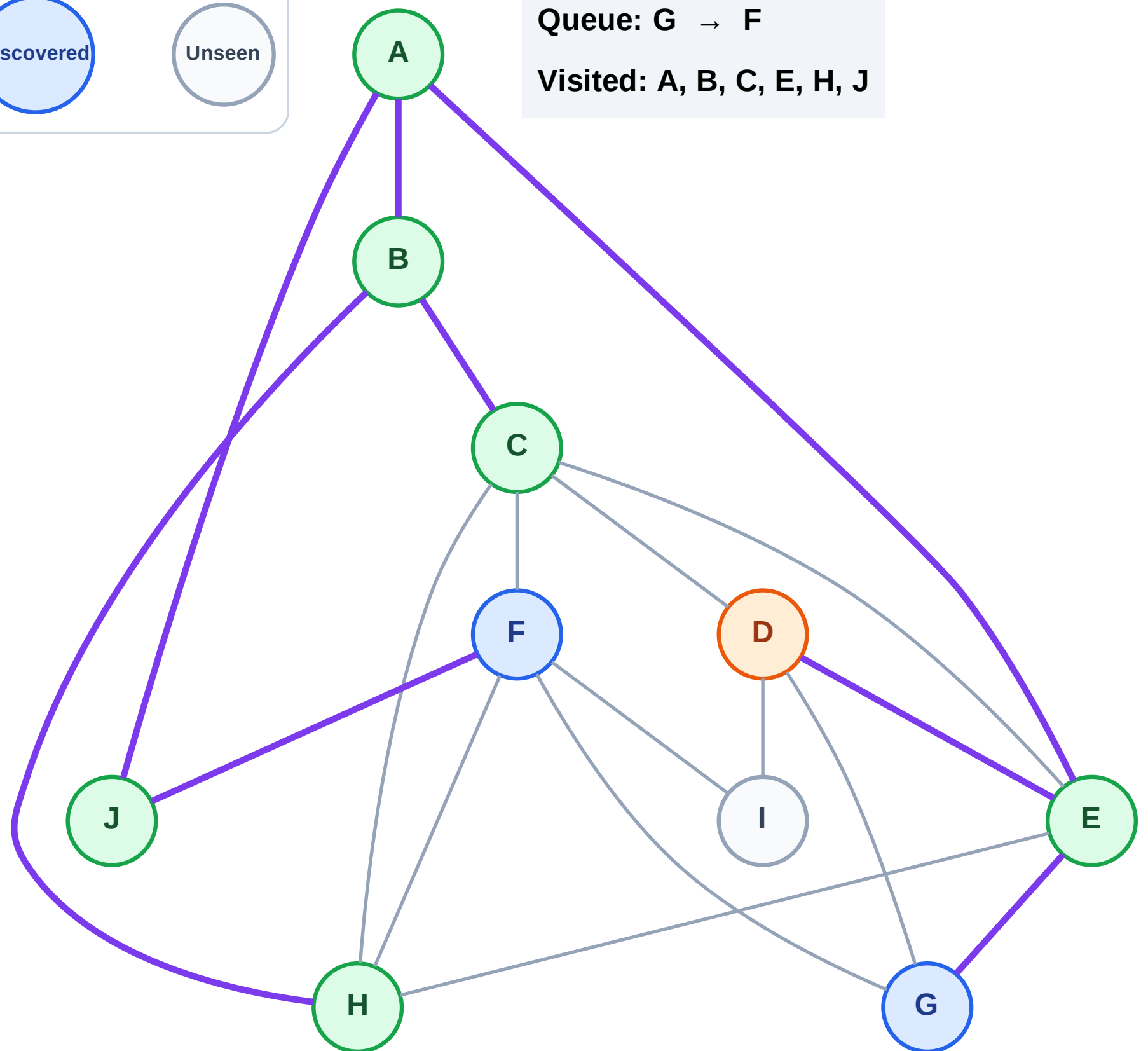
Step 14: Process node D

Legend



Queue: G → F

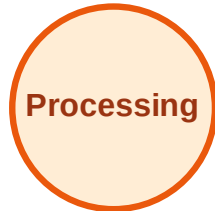
Visited: A, B, C, E, H, J



BFS Traversal

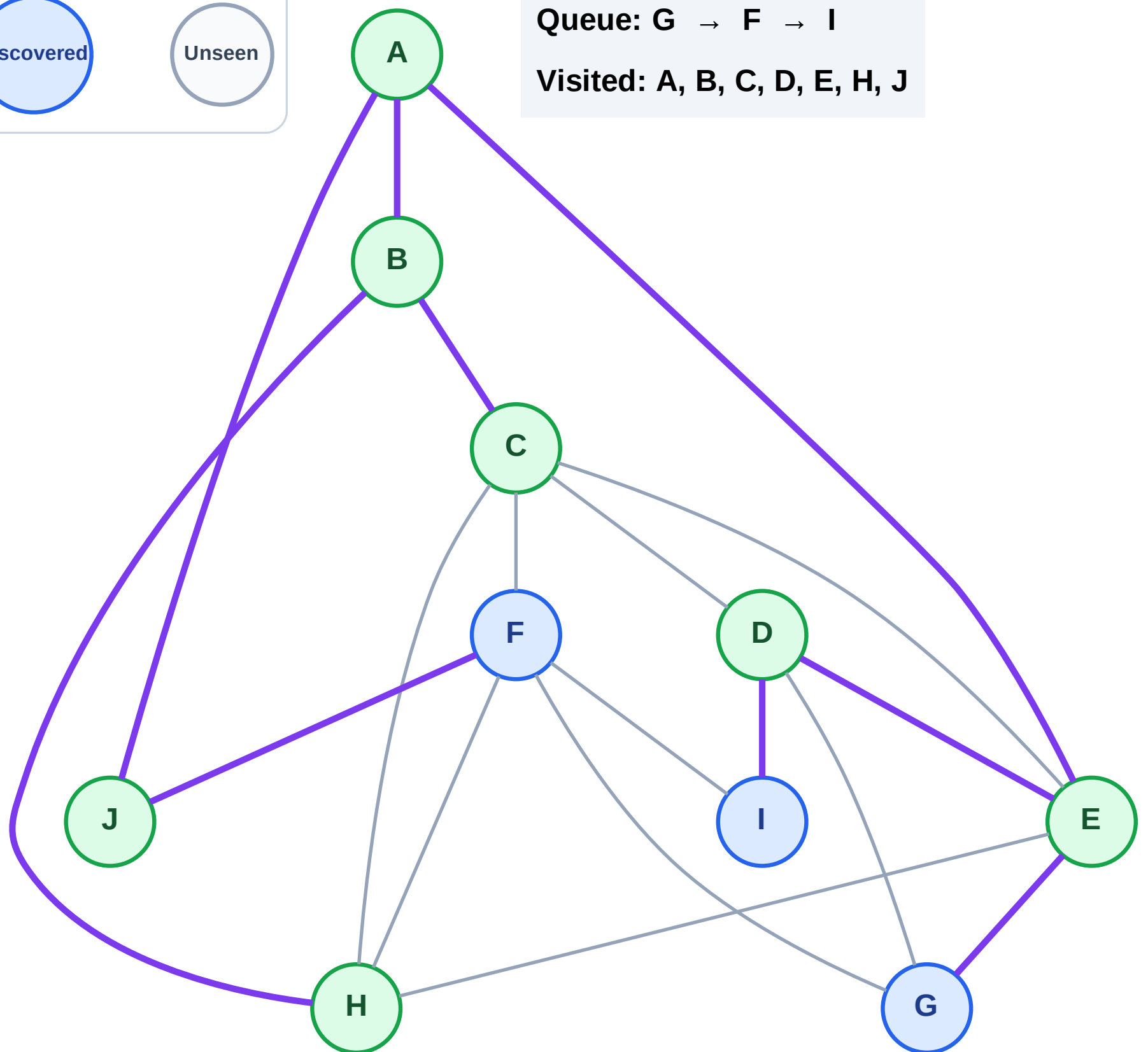
Step 15: Discover I from D

Legend

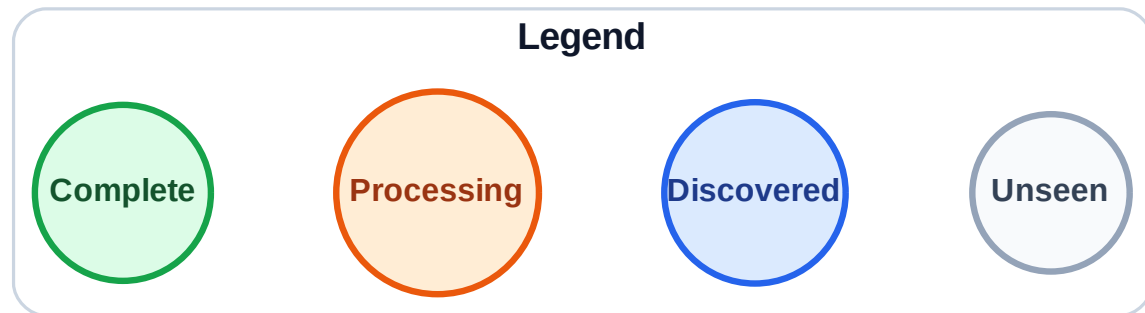


Queue: G → F → I

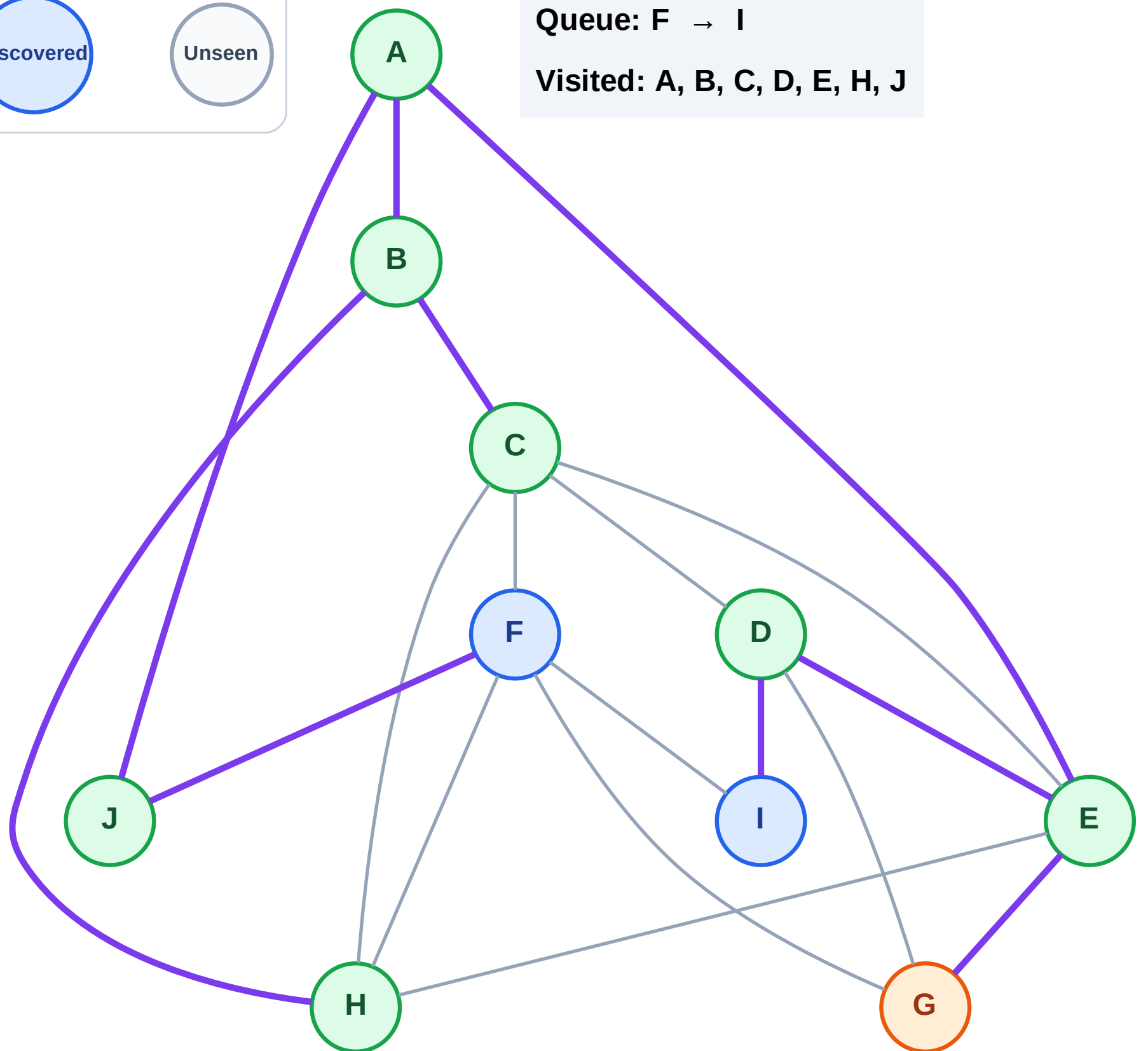
Visited: A, B, C, D, E, H, J



Step 16: Process node G



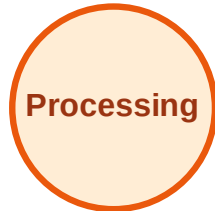
Visited: A, B, C, D, E, H, J



BFS Traversal

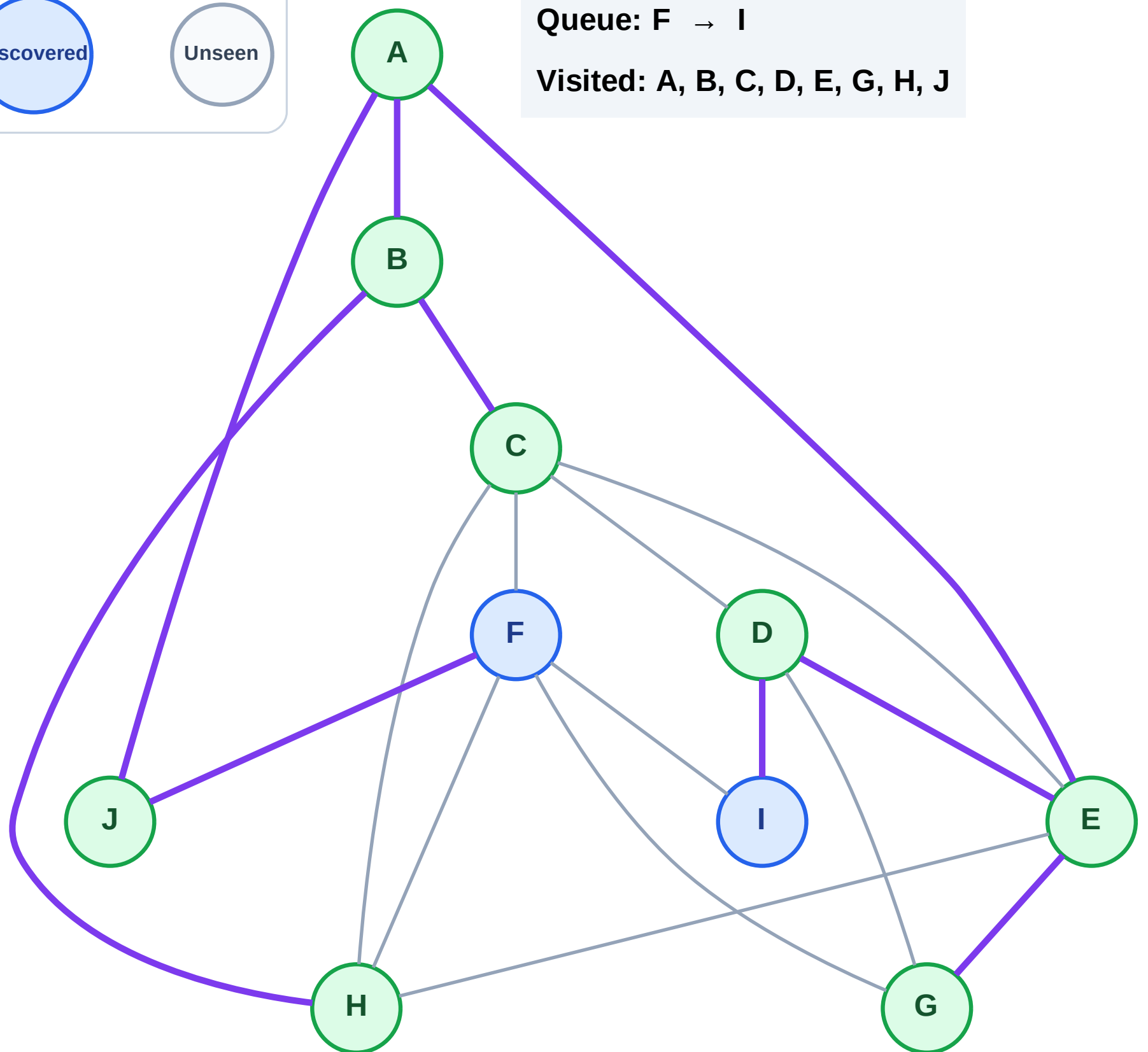
Step 17: No new neighbors from G

Legend



Queue: F → I

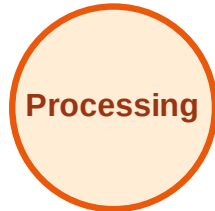
Visited: A, B, C, D, E, G, H, J



BFS Traversal

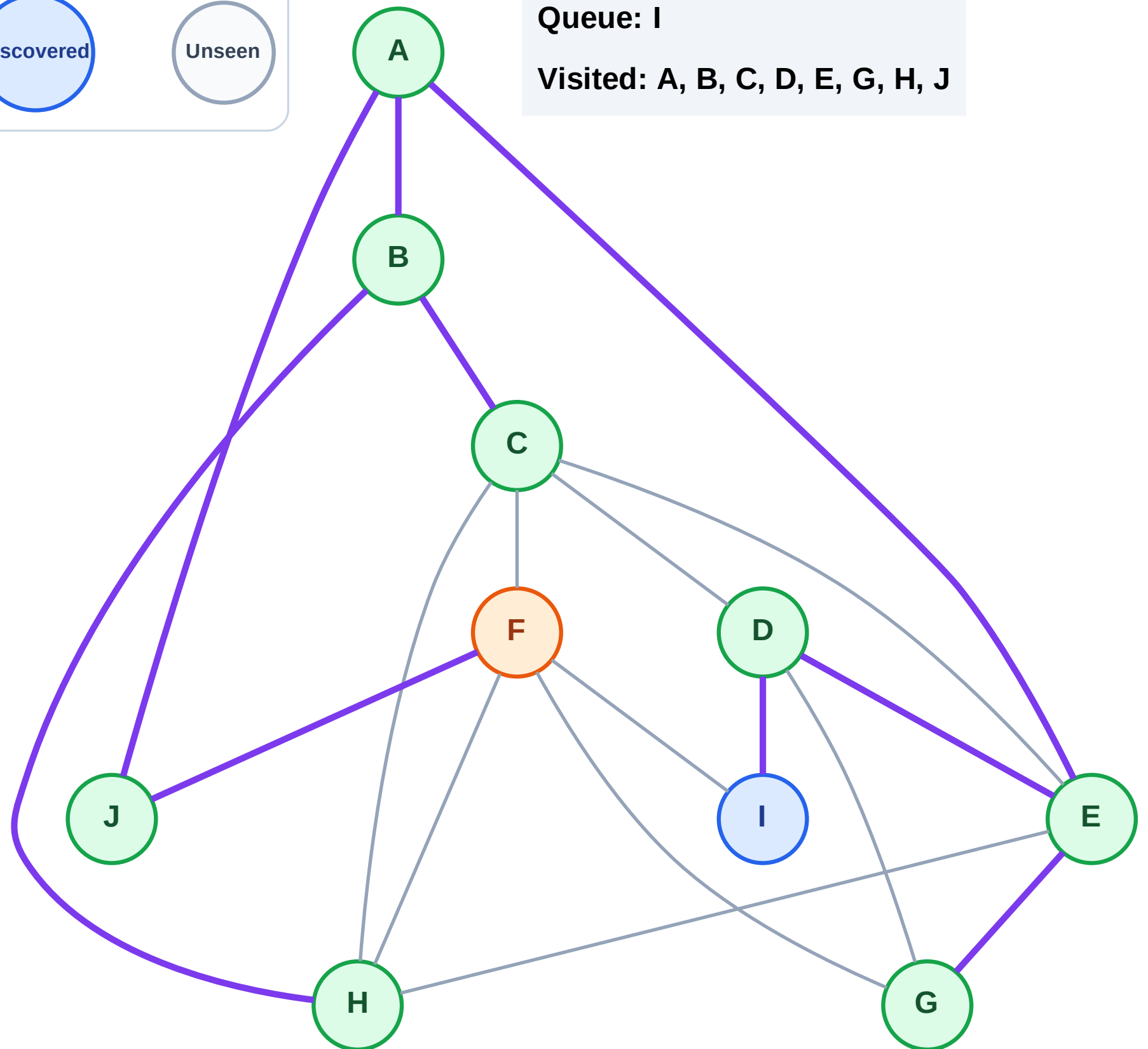
Step 18: Process node F

Legend



Queue: I

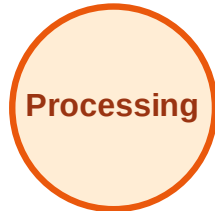
Visited: A, B, C, D, E, G, H, J



BFS Traversal

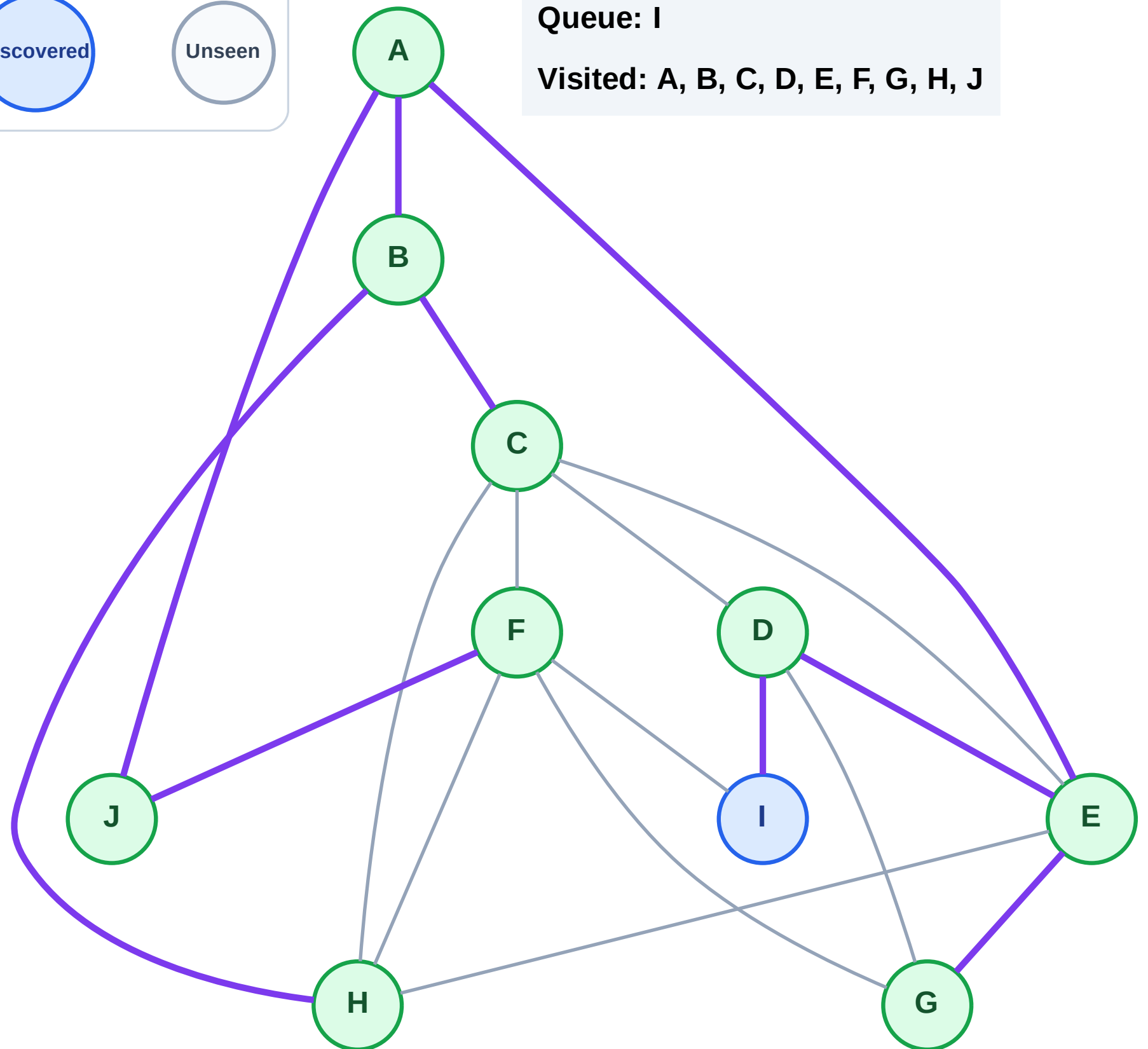
Step 19: No new neighbors from F

Legend



Queue: I

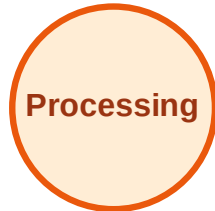
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

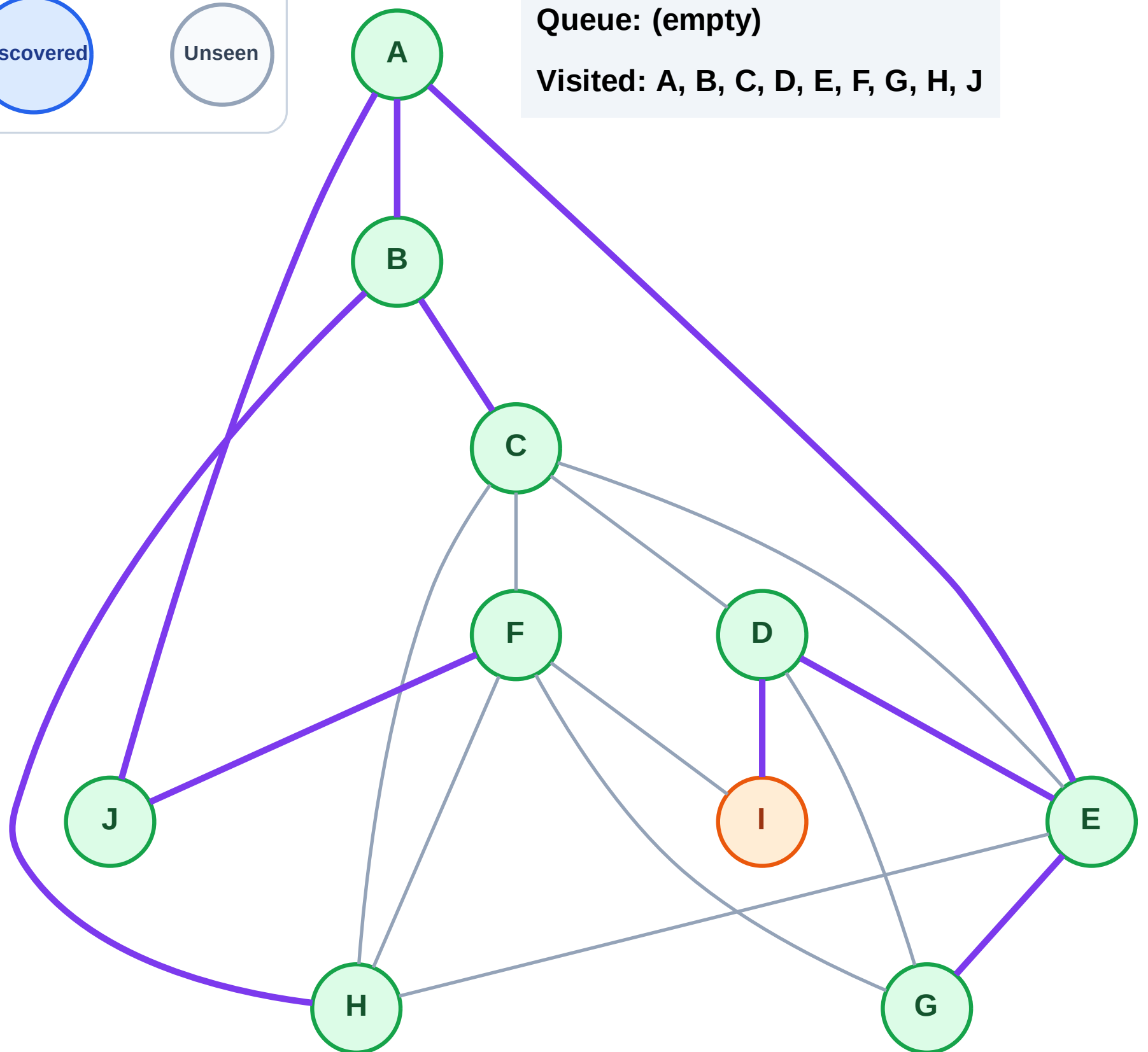
Step 20: Process node I

Legend



Queue: (empty)

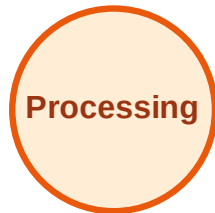
Visited: A, B, C, D, E, F, G, H, J



BFS Traversal

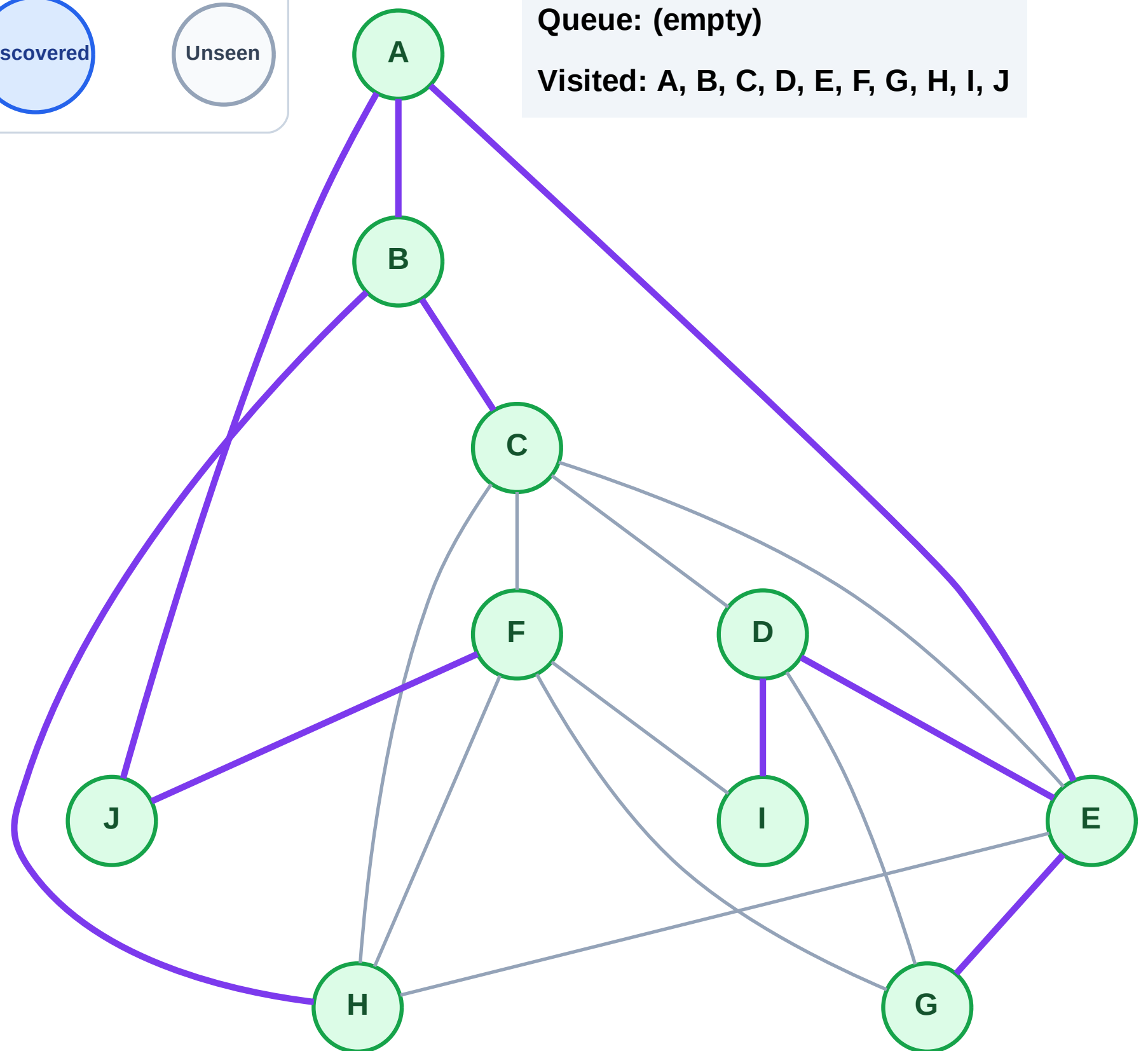
Step 21: No new neighbors from I

Legend



Queue: (empty)

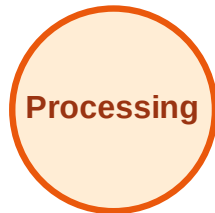
Visited: A, B, C, D, E, F, G, H, I, J



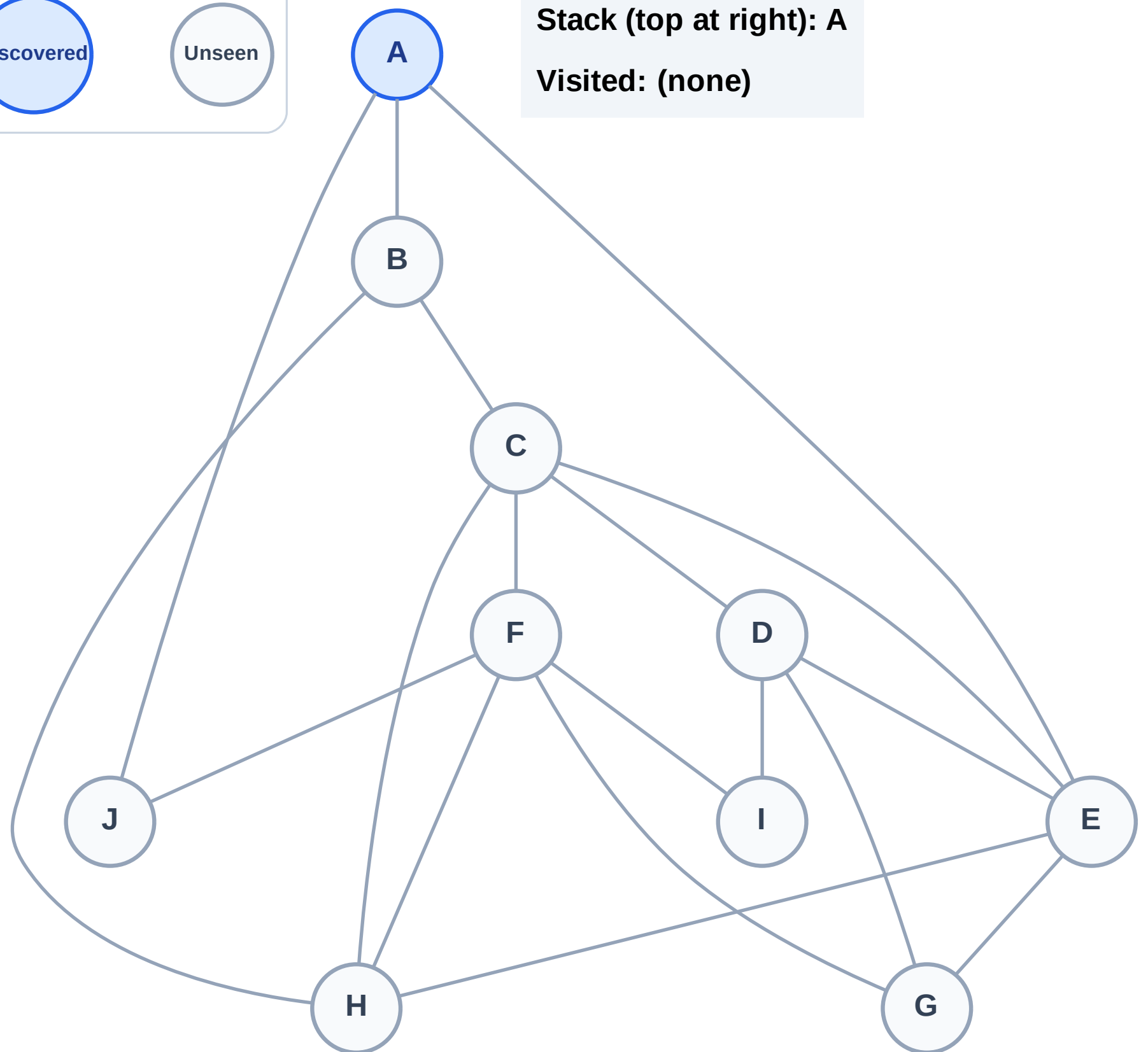
DFS Traversal

Step 1: Start at A

Legend



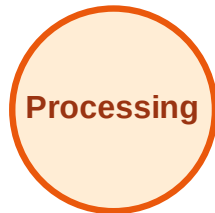
Stack (top at right): A
Visited: (none)



DFS Traversal

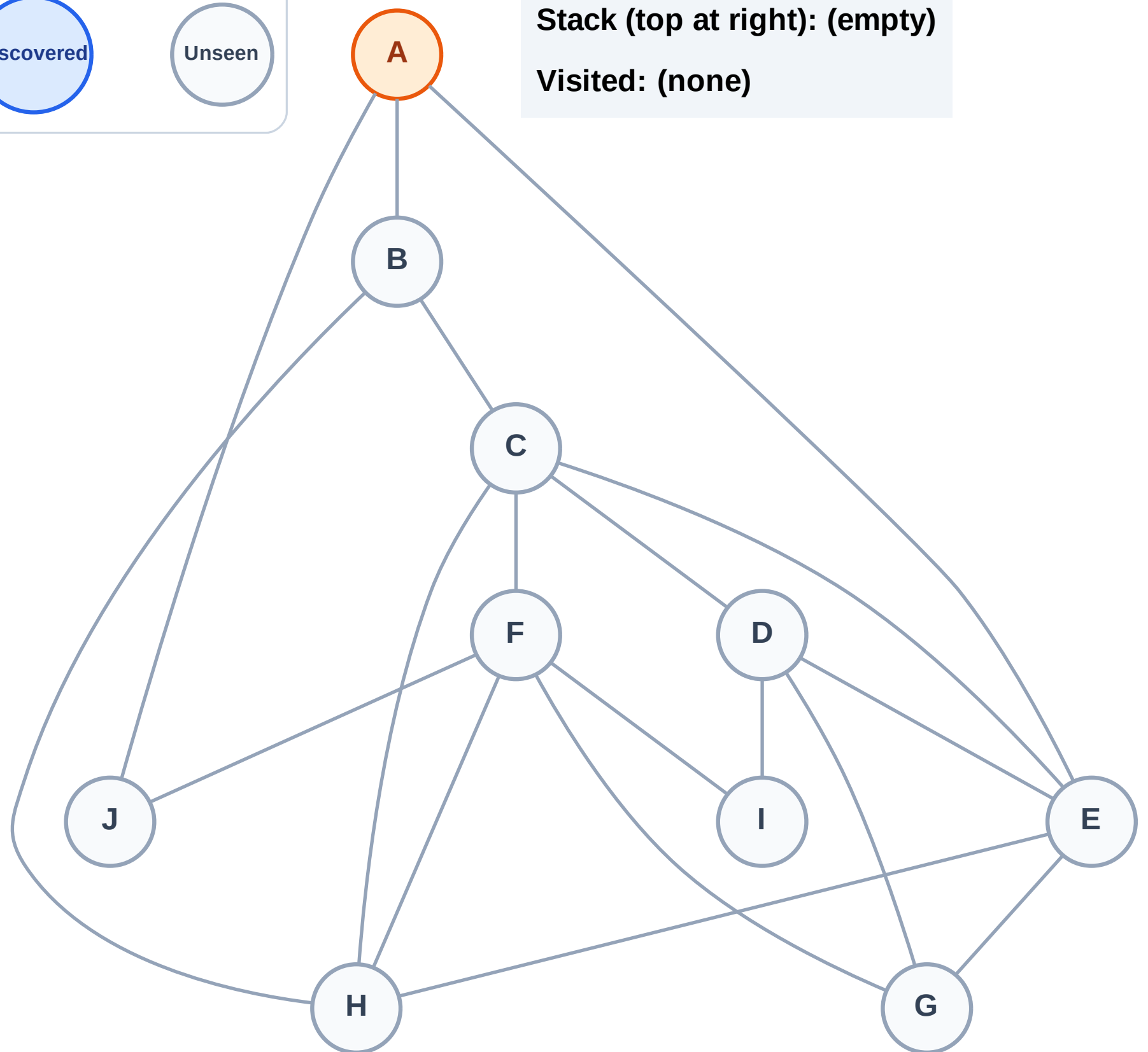
Step 2: Process node A

Legend



Stack (top at right): (empty)

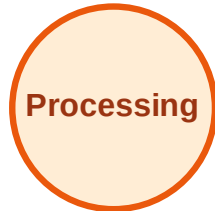
Visited: (none)



DFS Traversal

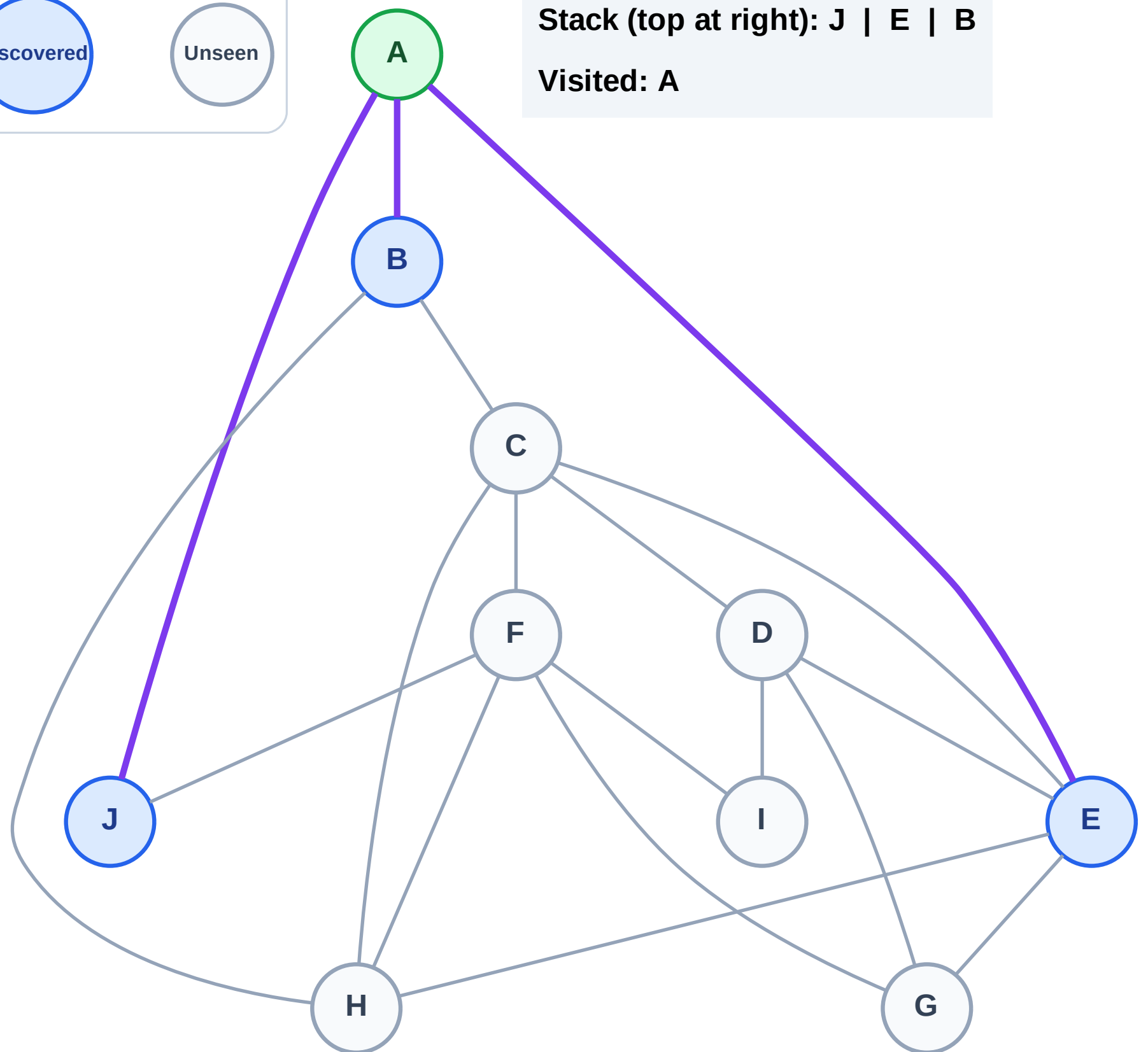
Step 3: Discover J, E, B from A

Legend



Stack (top at right): J | E | B

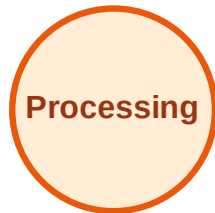
Visited: A



DFS Traversal

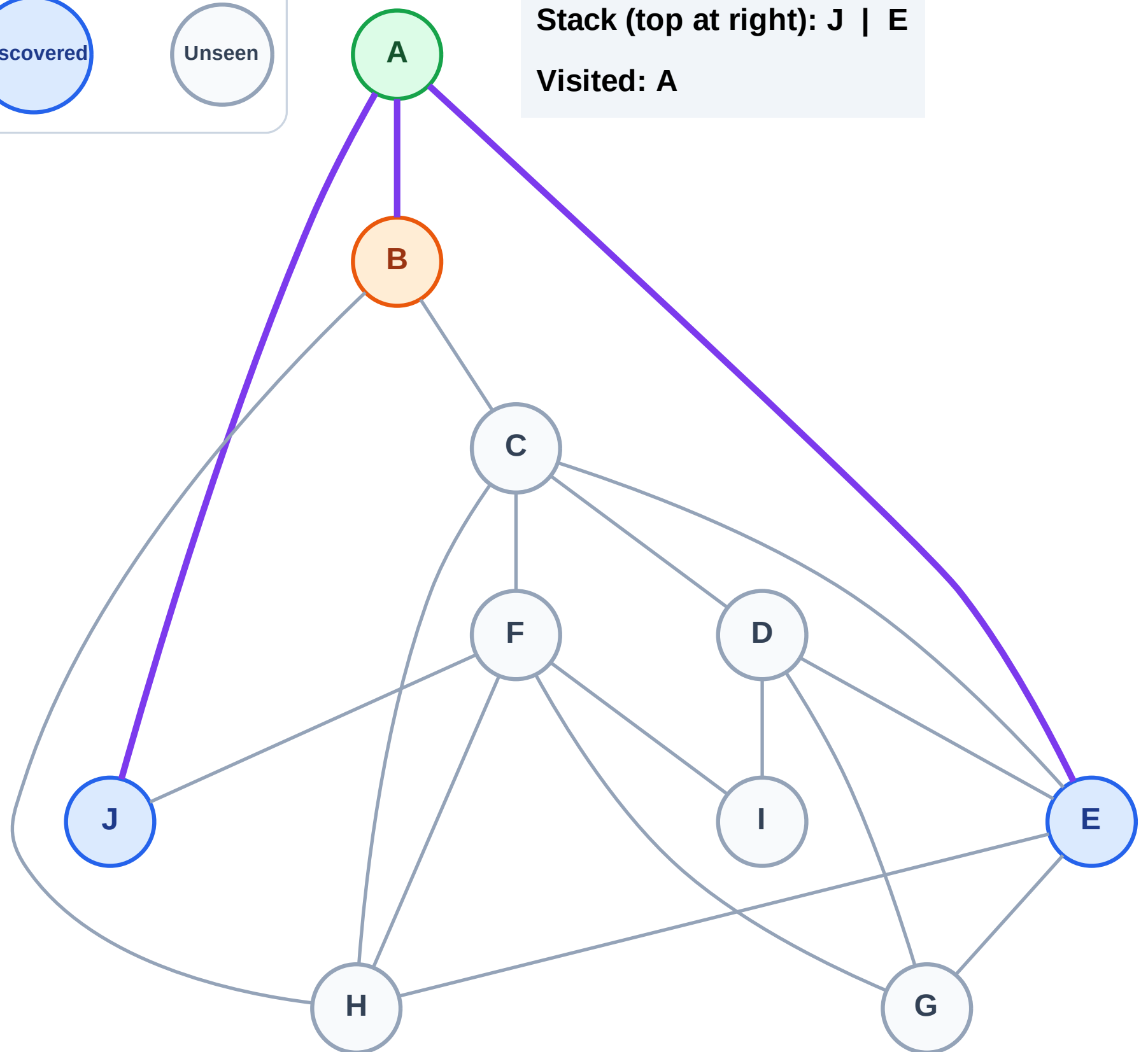
Step 4: Process node B

Legend



Stack (top at right): J | E

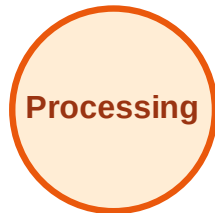
Visited: A



DFS Traversal

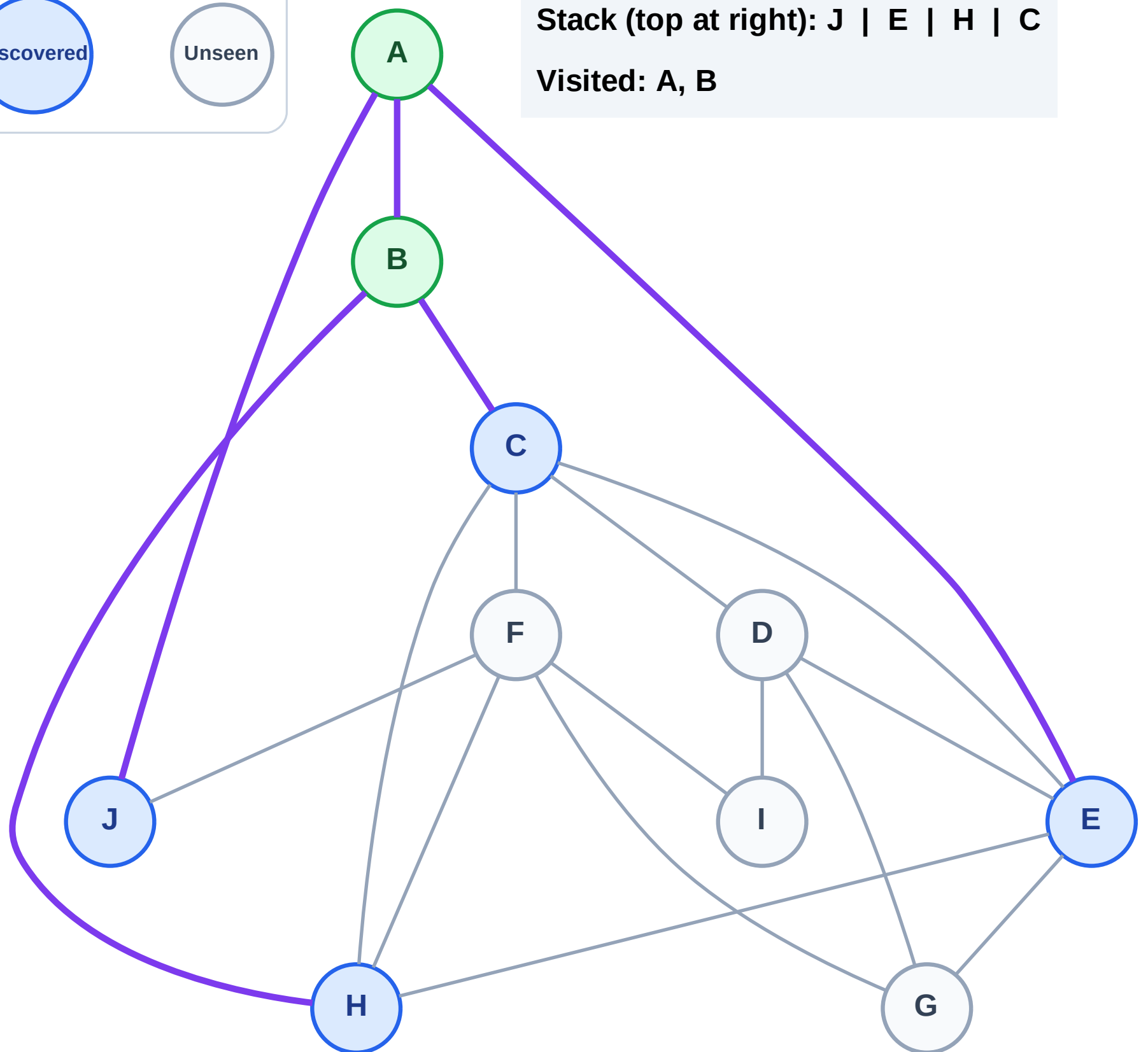
Step 5: Discover H, C from B

Legend



Stack (top at right): J | E | H | C

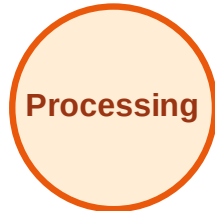
Visited: A, B



DFS Traversal

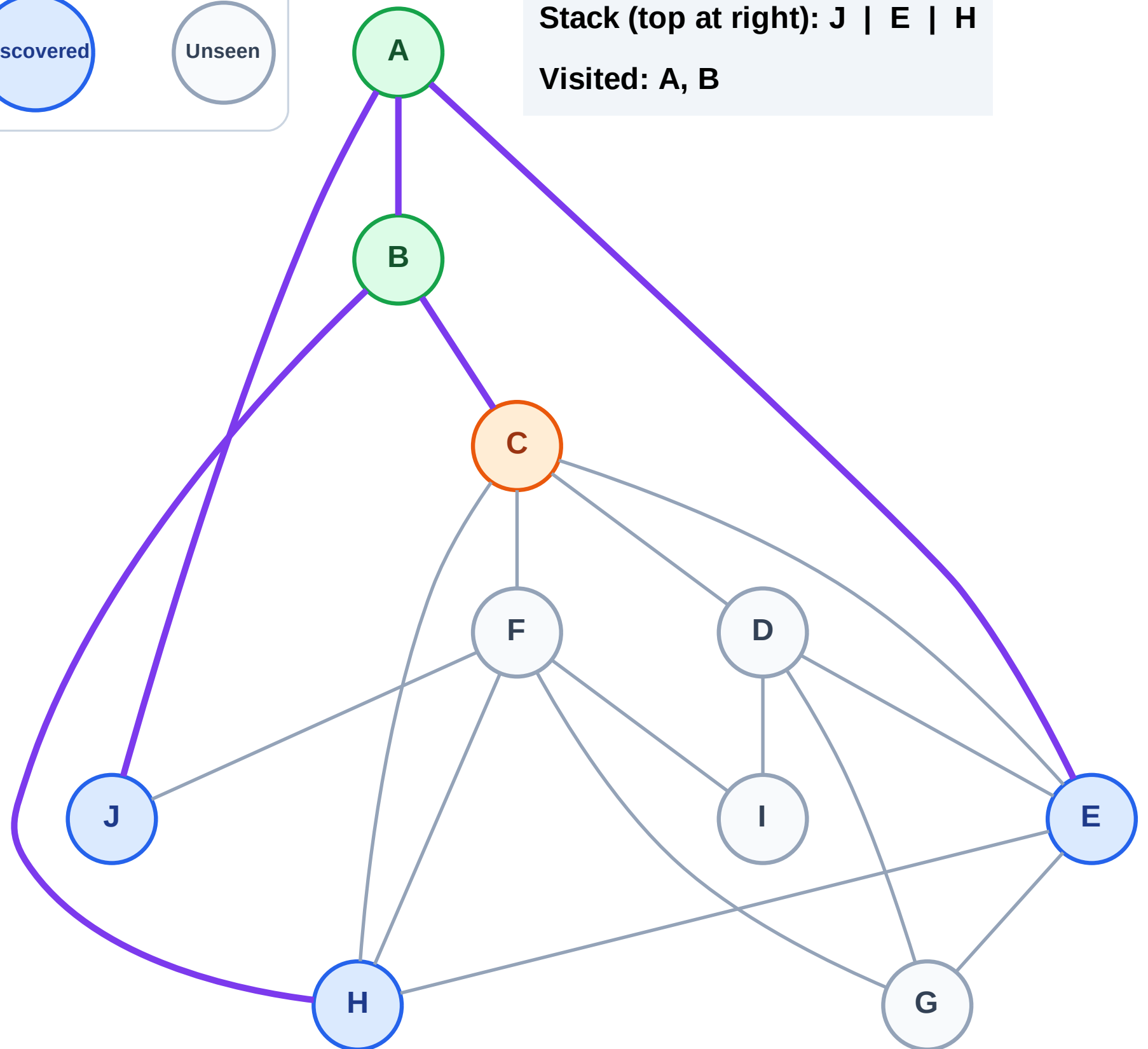
Step 6: Process node C

Legend



Stack (top at right): J | E | H

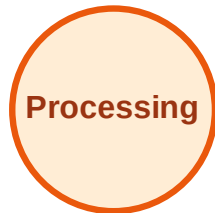
Visited: A, B



DFS Traversal

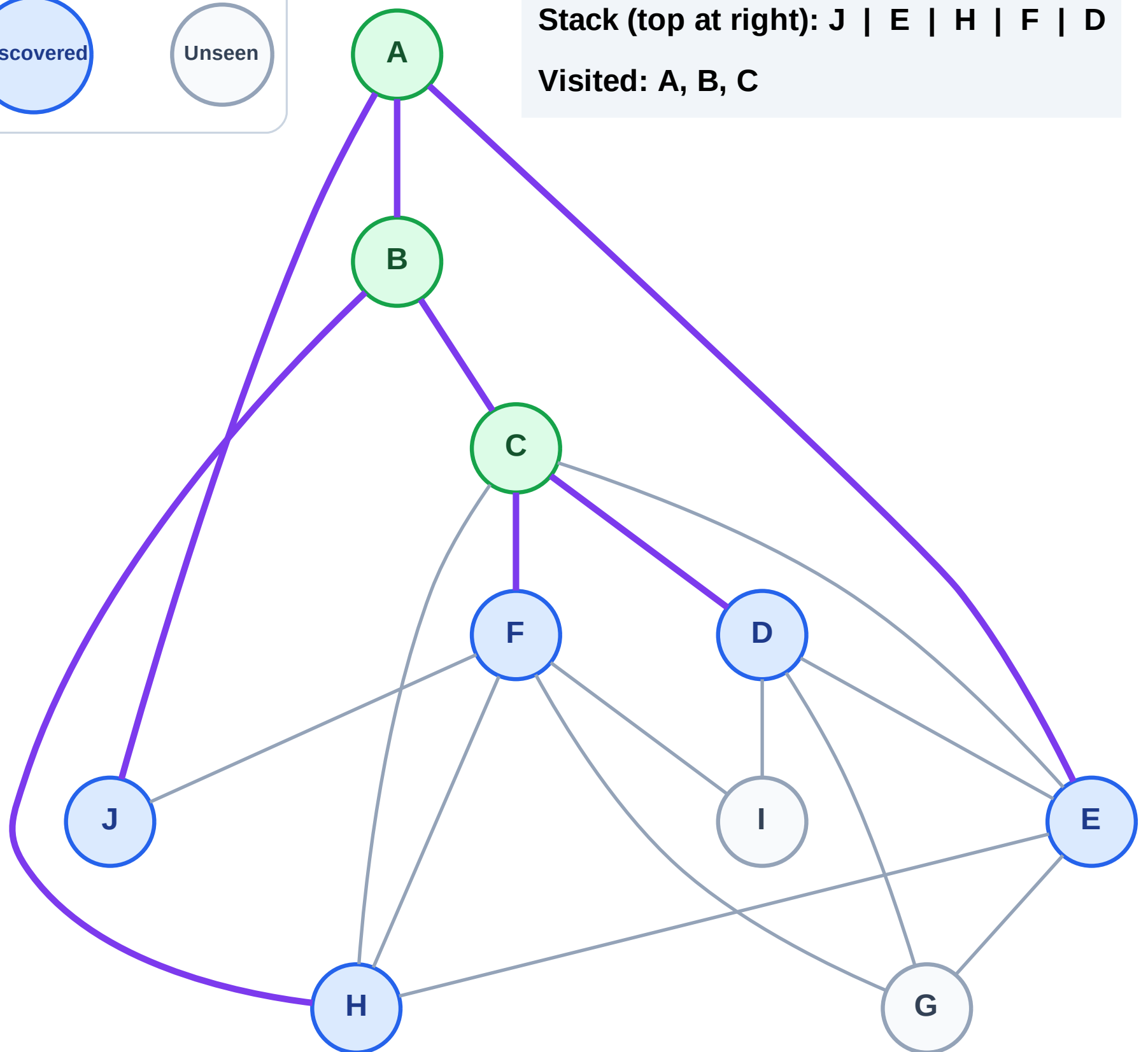
Step 7: Discover F, D from C

Legend



Stack (top at right): J | E | H | F | D

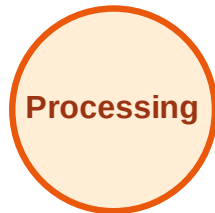
Visited: A, B, C



DFS Traversal

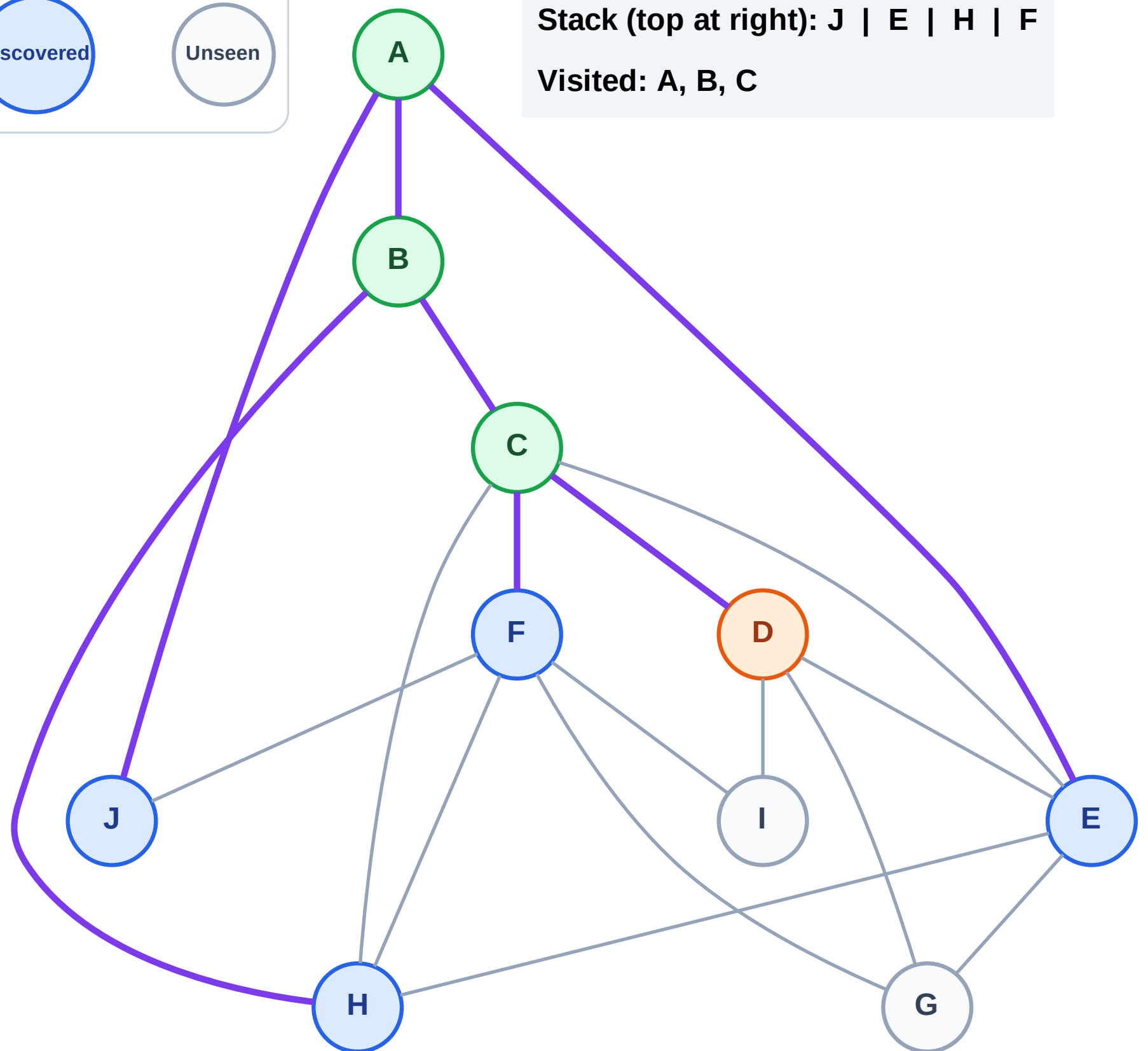
Step 8: Process node D

Legend



Stack (top at right): J | E | H | F

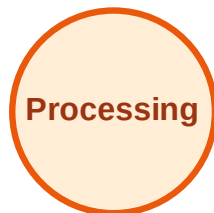
Visited: A, B, C



DFS Traversal

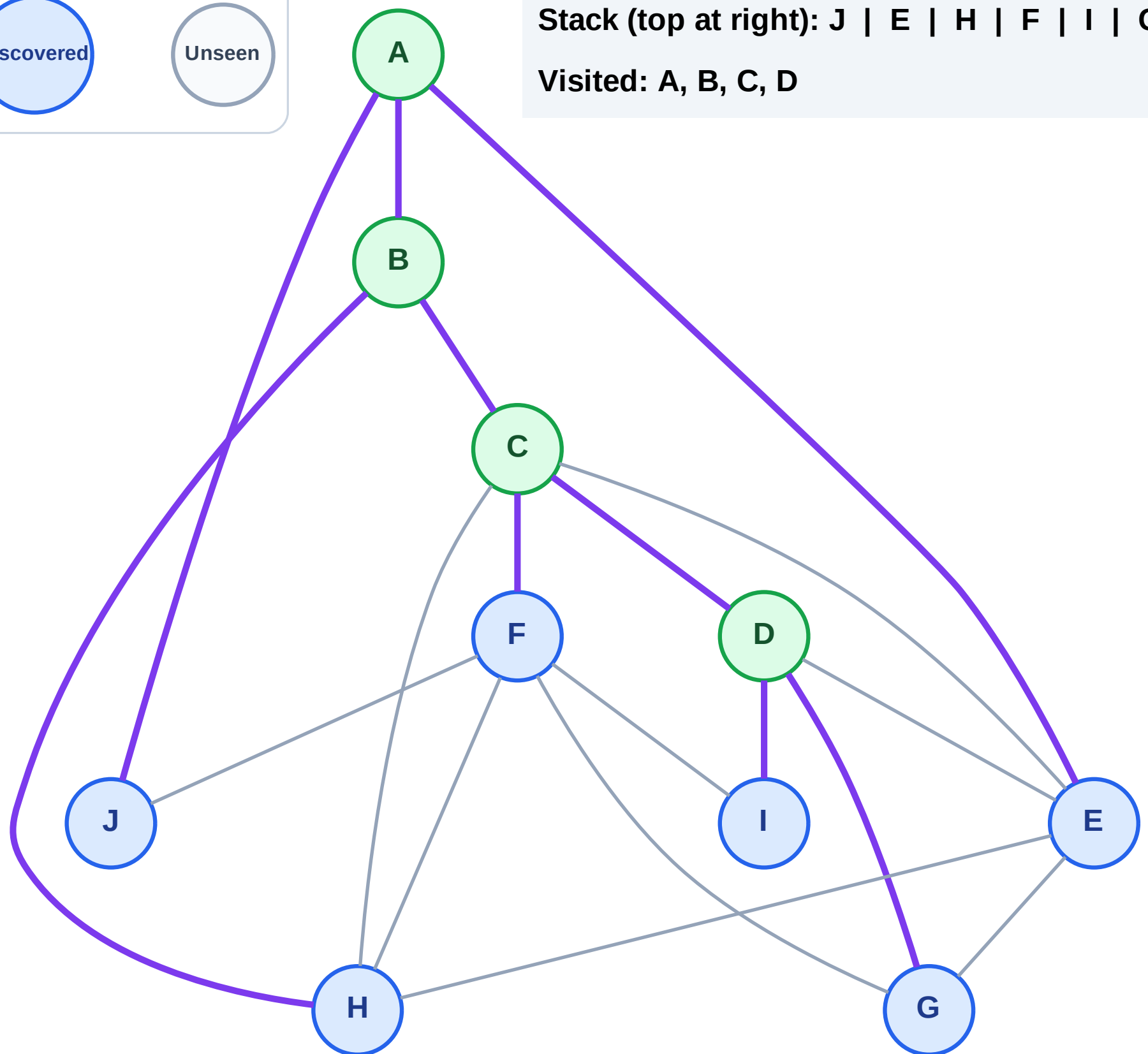
Step 9: Discover I, G from D

Legend



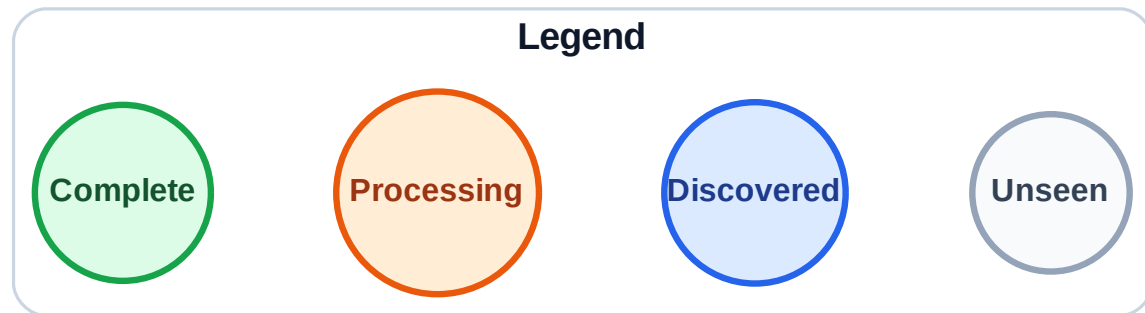
Stack (top at right): J | E | H | F | I | G

Visited: A, B, C, D



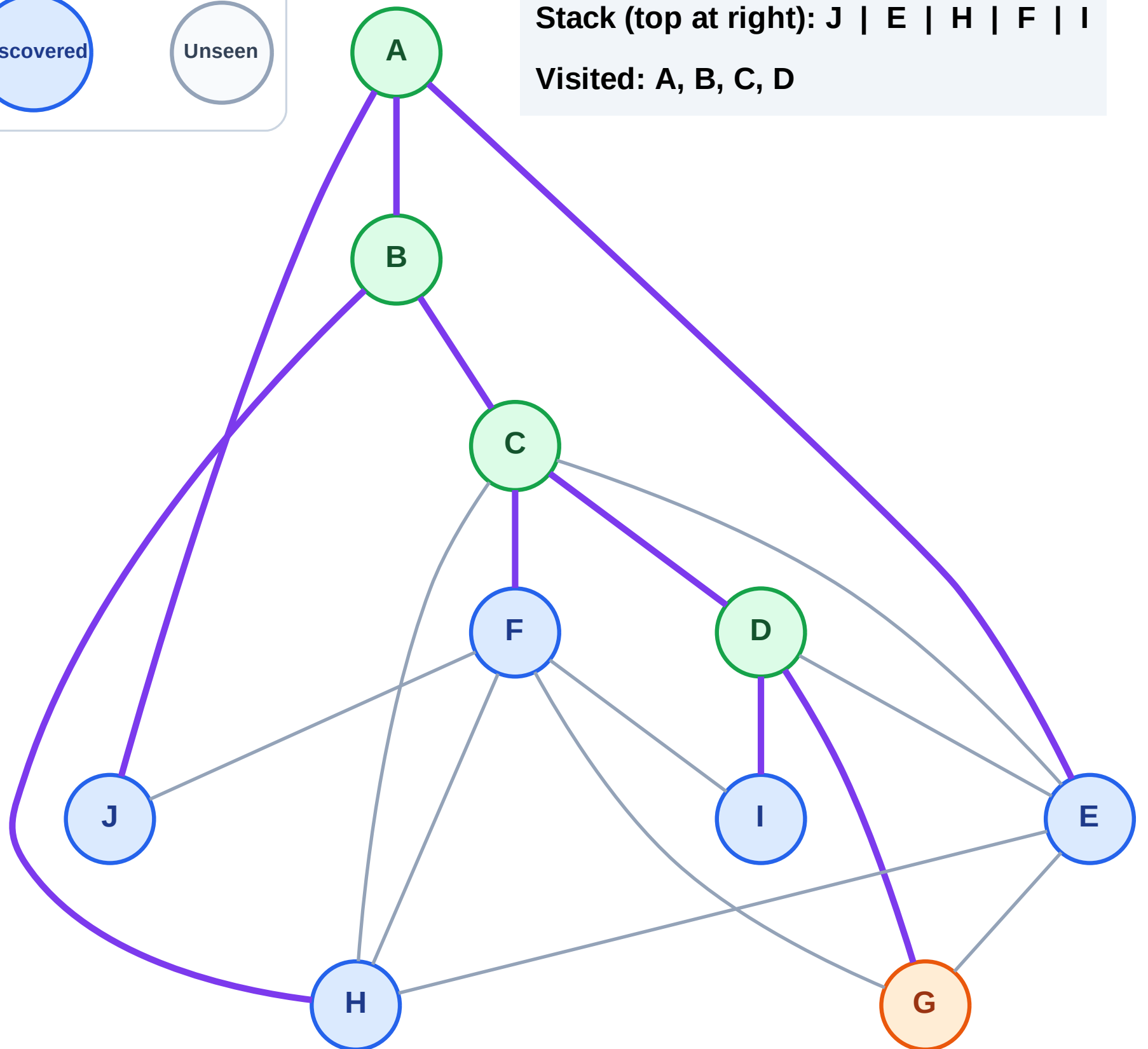
Step 10: Process node G

Step 10: Process node G



Stack (top at right): J | E | H | F | I

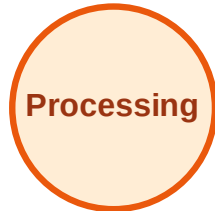
Visited: A, B, C, D



DFS Traversal

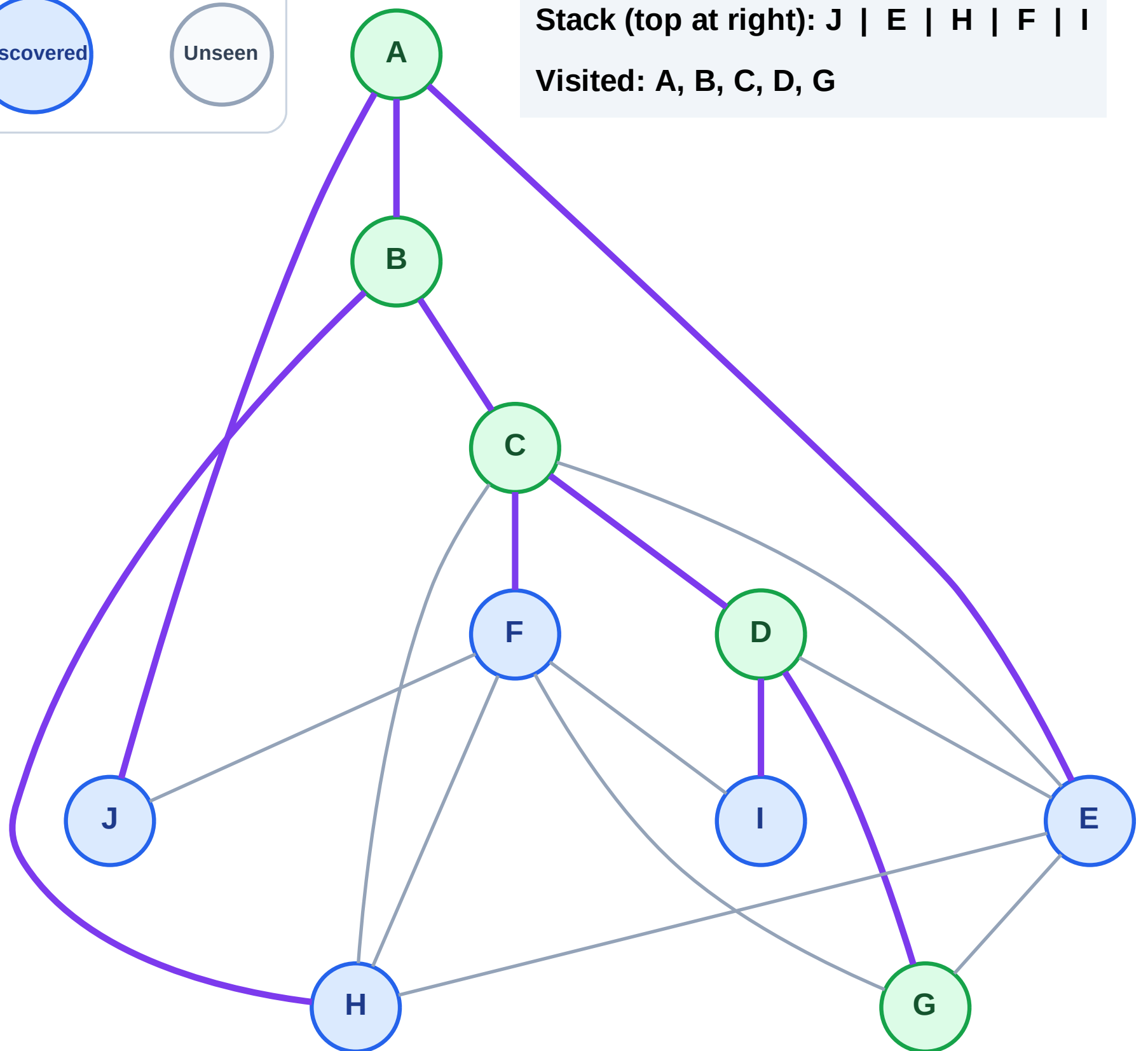
Step 11: No new neighbors from G

Legend



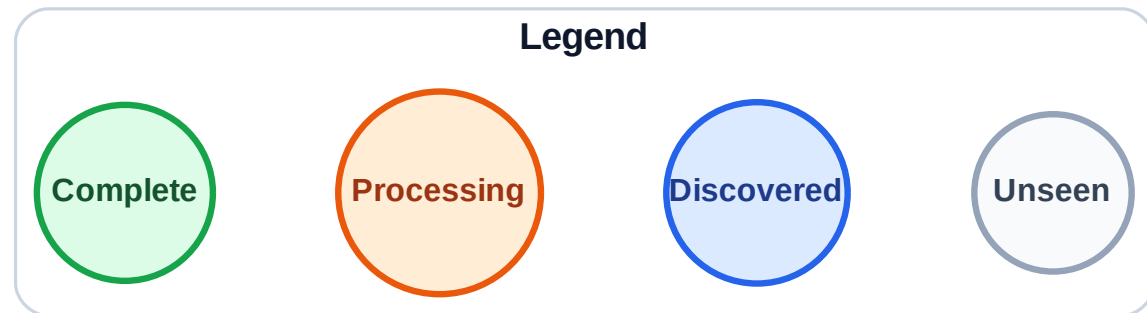
Stack (top at right): J | E | H | F | I

Visited: A, B, C, D, G



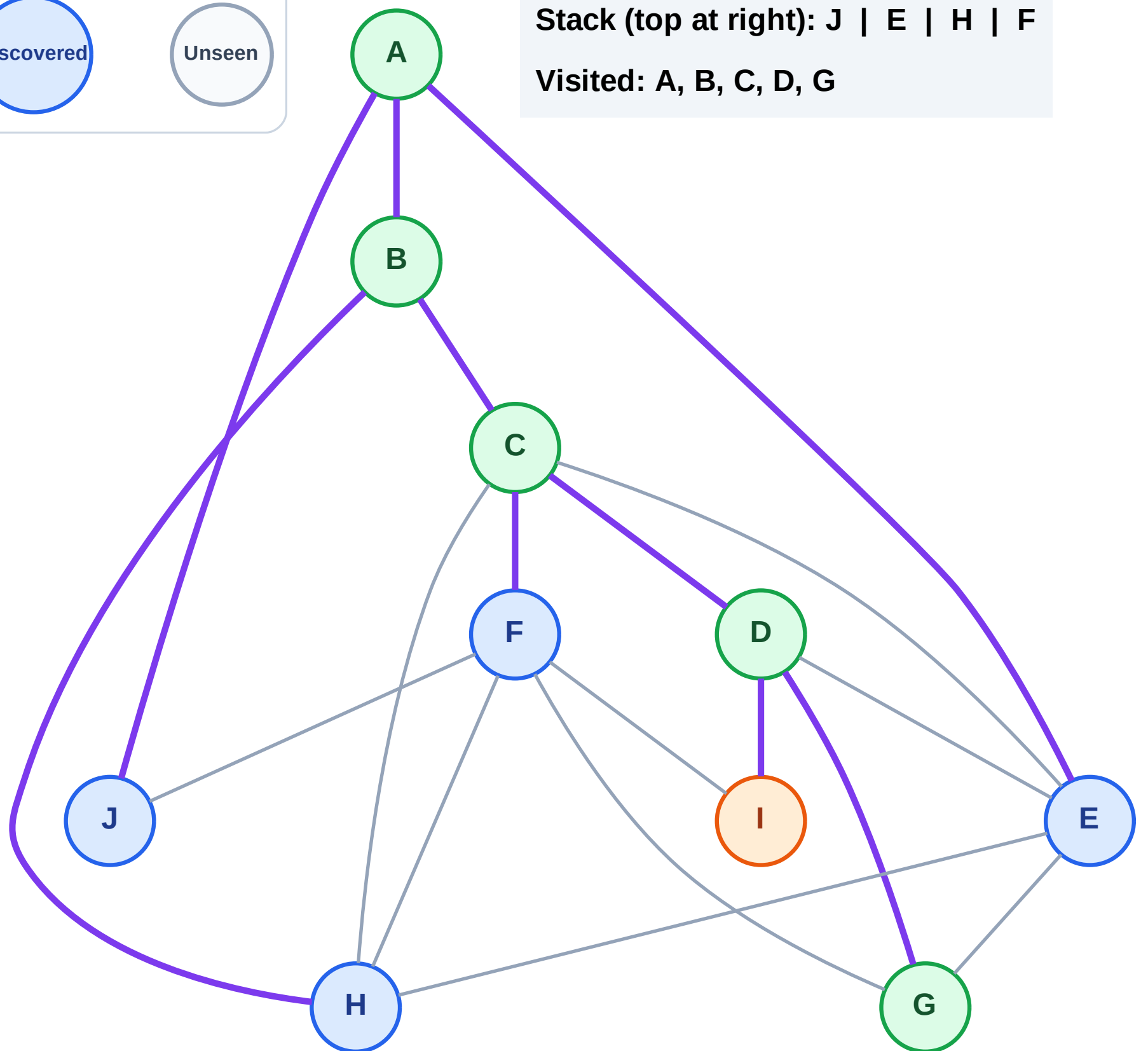
Step 12: Process node I

Step 12: Process node I



Stack (top at right): J | E | H | F

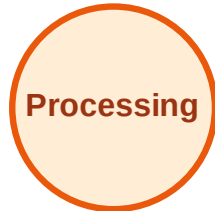
Visited: A, B, C, D, G



DFS Traversal

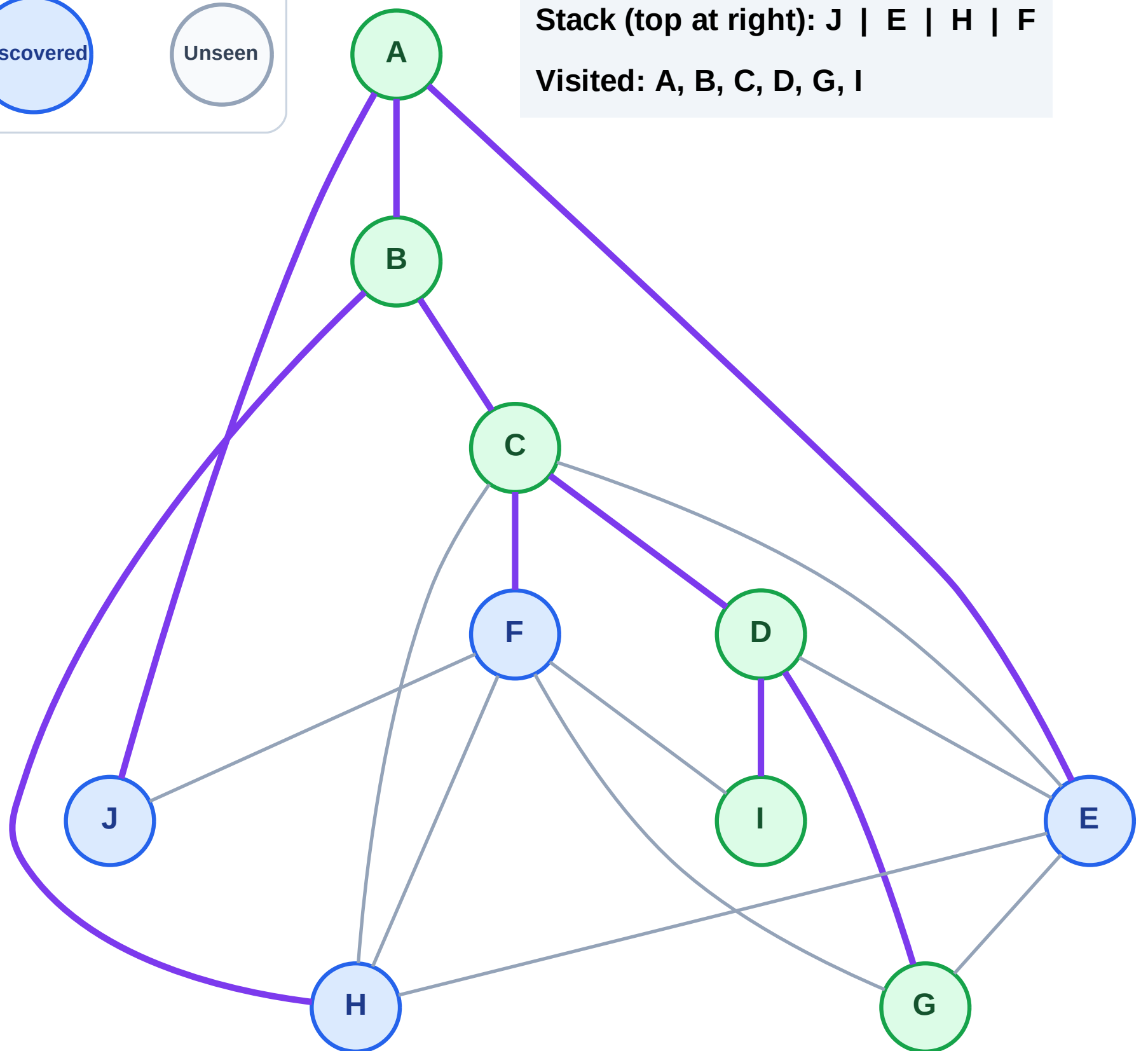
Step 13: No new neighbors from I

Legend



Stack (top at right): J | E | H | F

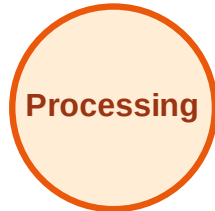
Visited: A, B, C, D, G, I



DFS Traversal

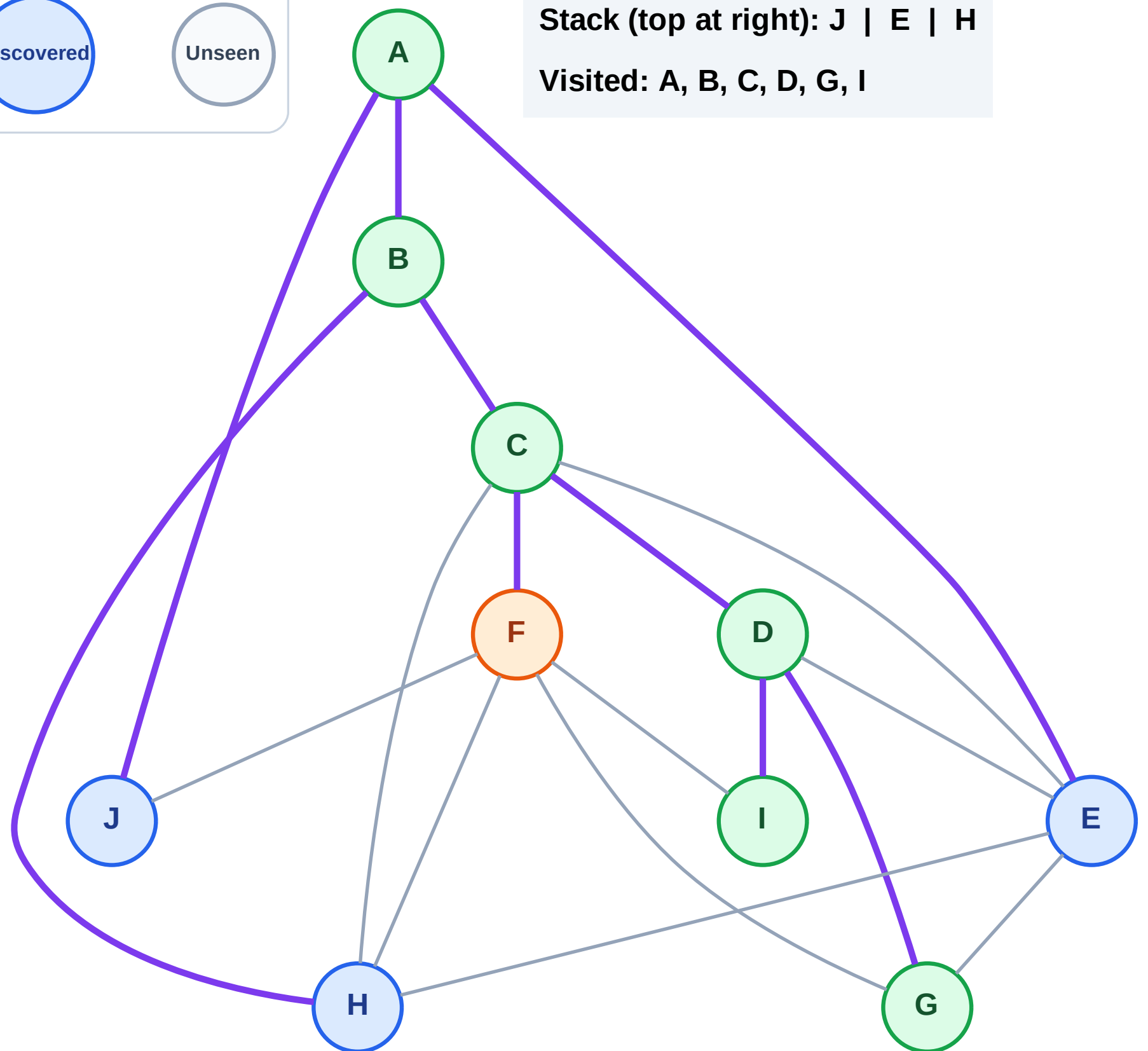
Step 14: Process node F

Legend



Stack (top at right): J | E | H

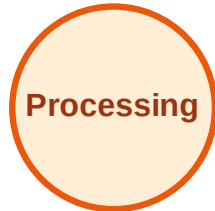
Visited: A, B, C, D, G, I



DFS Traversal

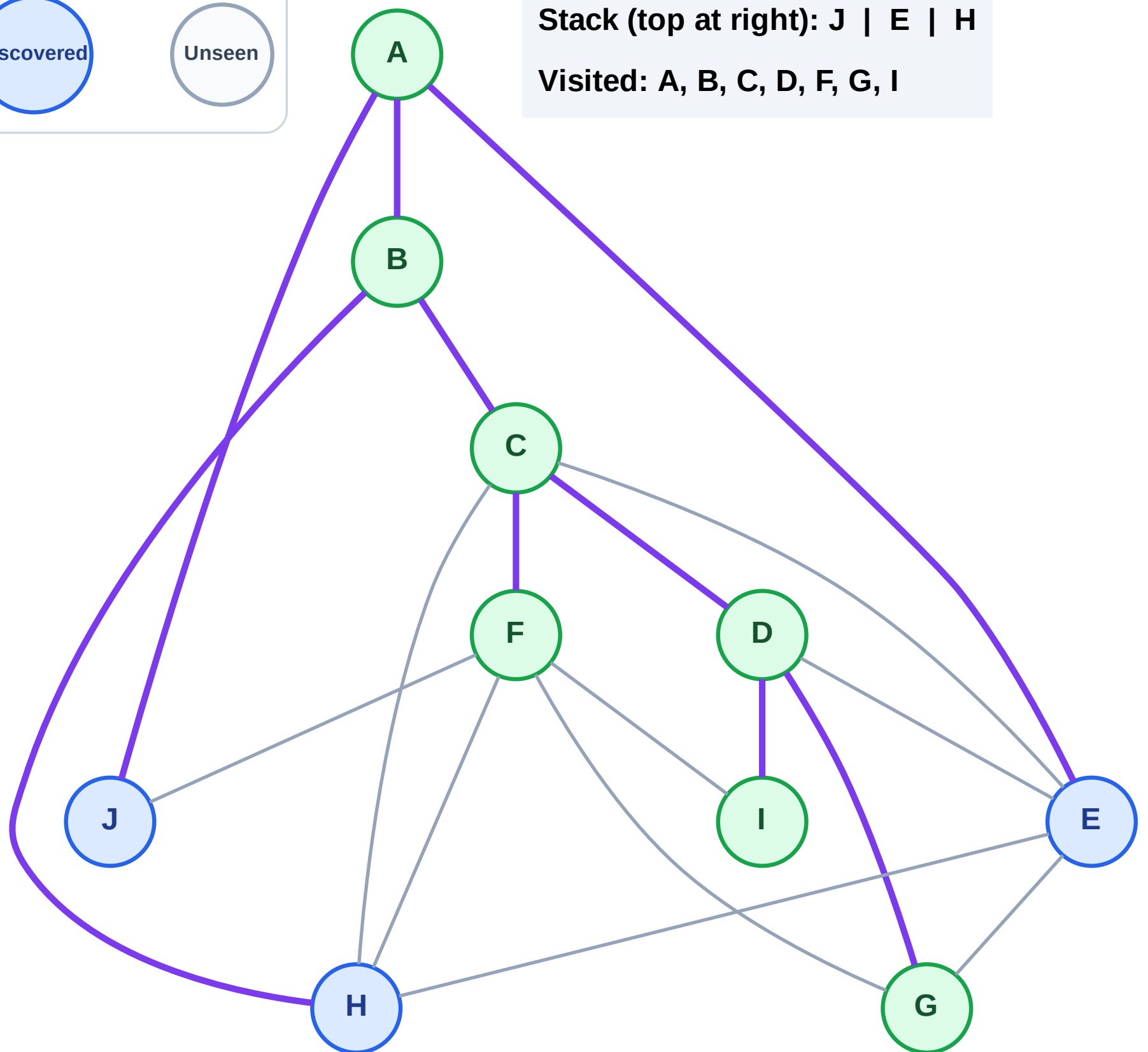
Step 15: No new neighbors from F

Legend



Stack (top at right): J | E | H

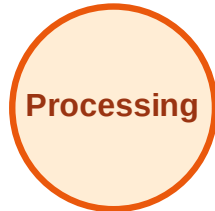
Visited: A, B, C, D, F, G, I



DFS Traversal

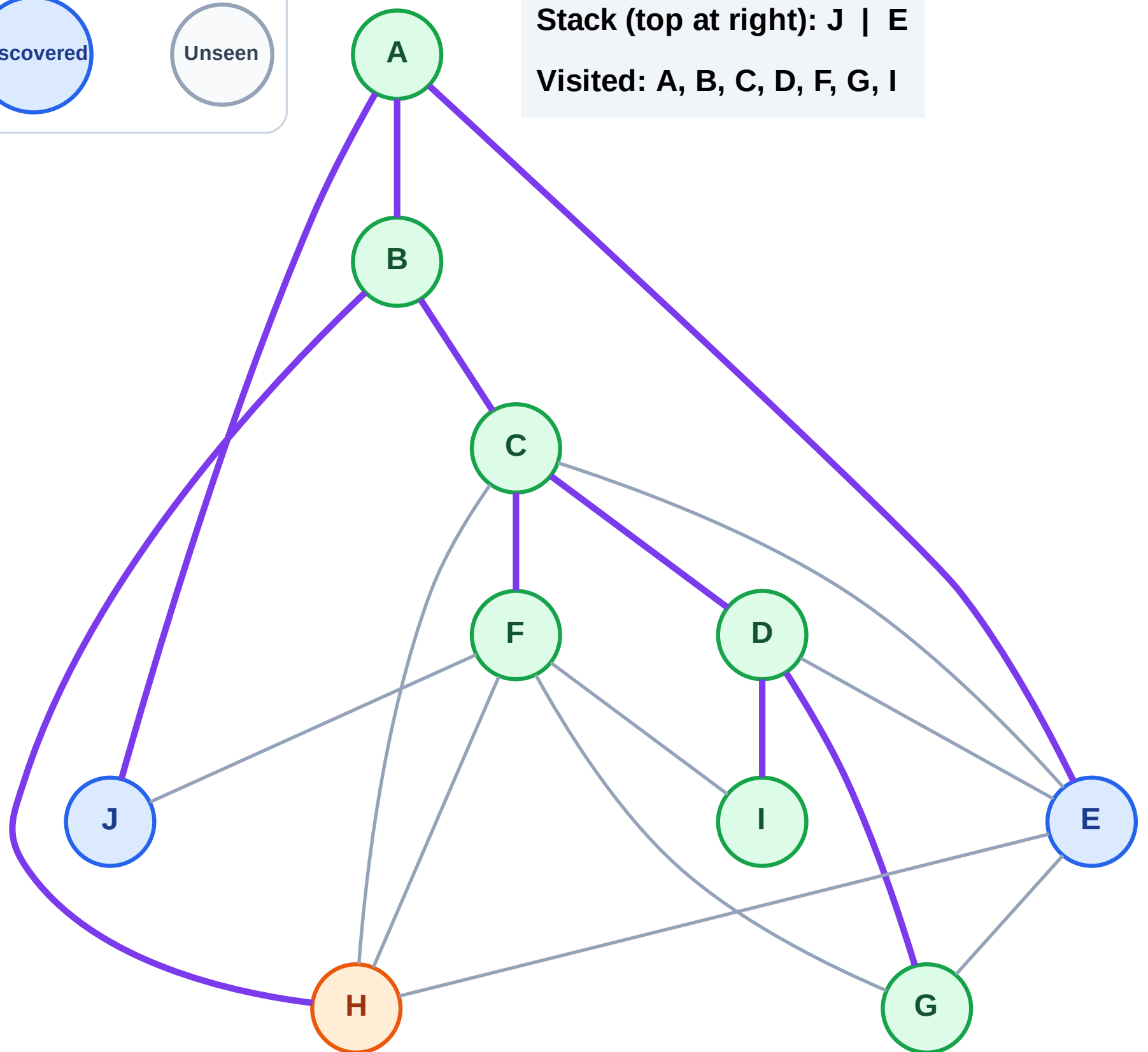
Step 16: Process node H

Legend



Stack (top at right): J | E

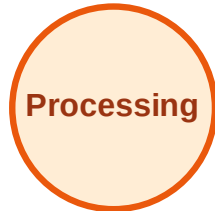
Visited: A, B, C, D, F, G, I



DFS Traversal

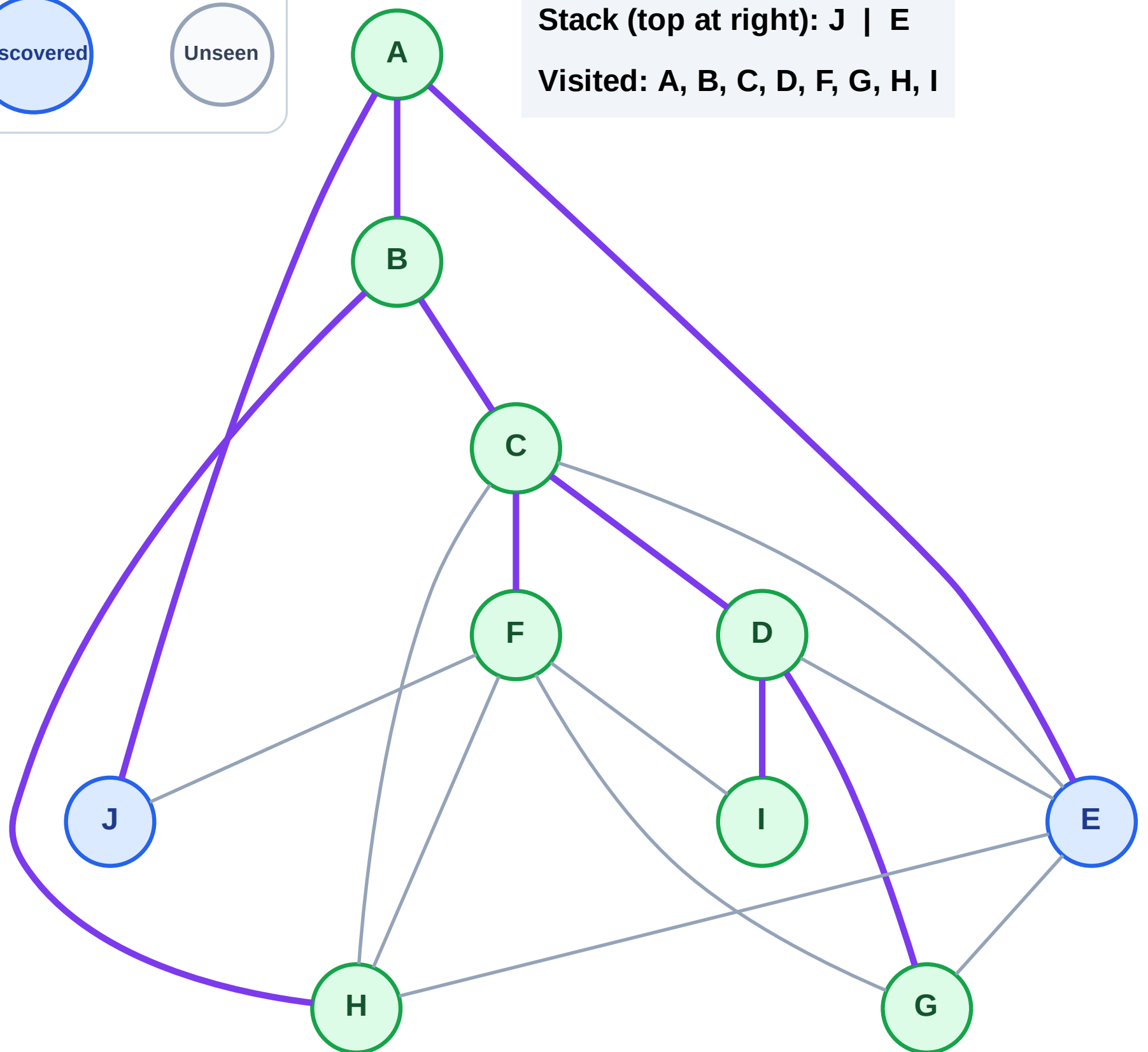
Step 17: No new neighbors from H

Legend



Stack (top at right): J | E

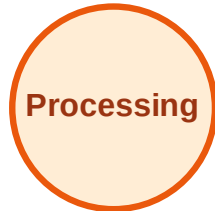
Visited: A, B, C, D, F, G, H, I



DFS Traversal

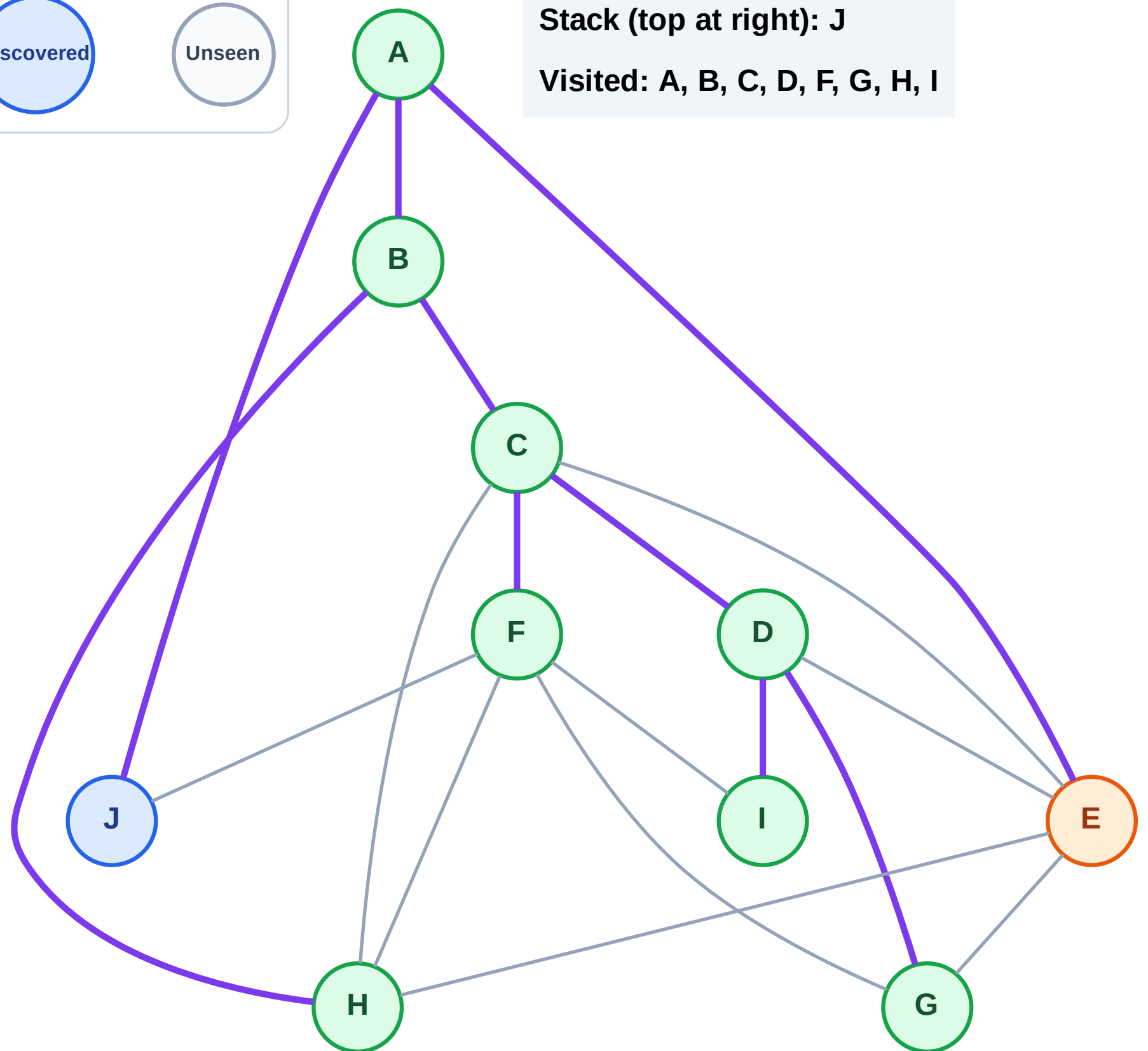
Step 18: Process node E

Legend



Stack (top at right): J

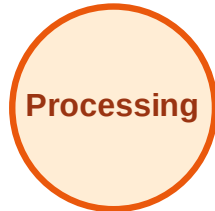
Visited: A, B, C, D, F, G, H, I



DFS Traversal

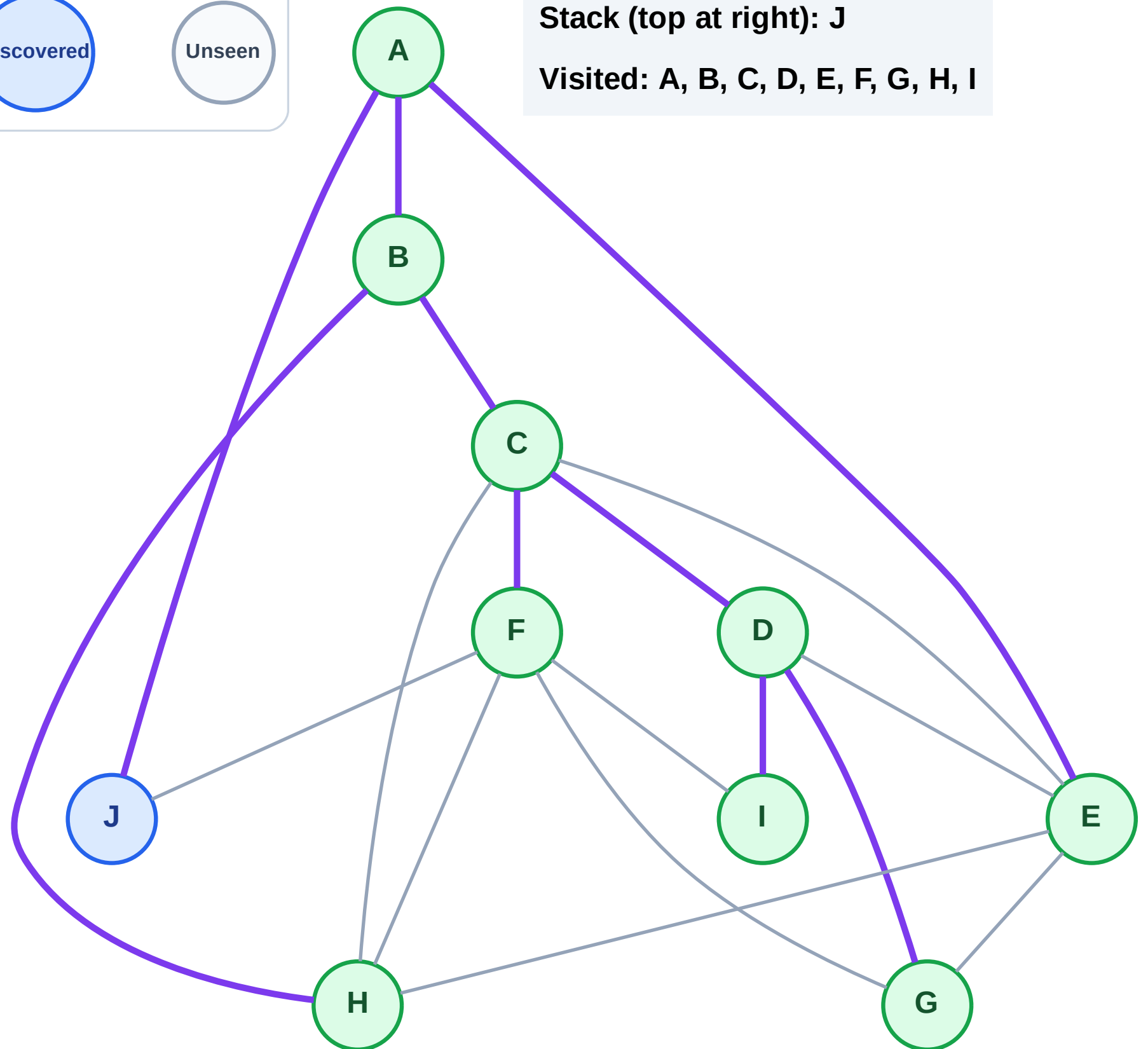
Step 19: No new neighbors from E

Legend



Stack (top at right): J

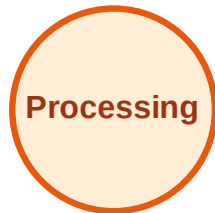
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

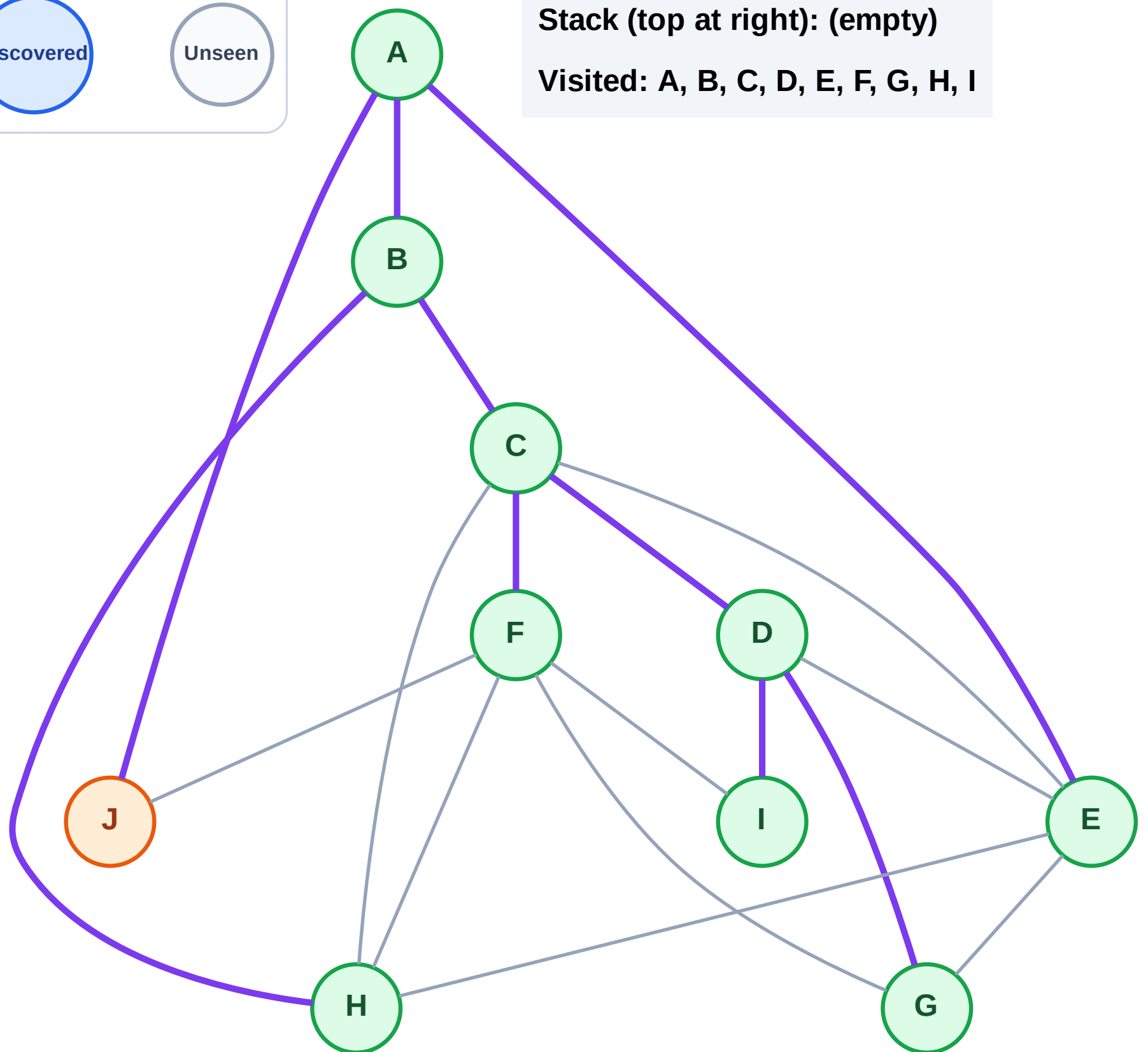
Step 20: Process node J

Legend



Stack (top at right): (empty)

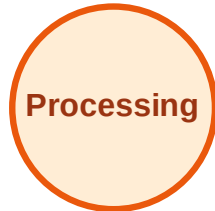
Visited: A, B, C, D, E, F, G, H, I



DFS Traversal

Step 21: No new neighbors from J

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

